Electronics

Nathan Warner

Northern Illinois
University

Computer Science
Northern Illinois University
United States

Contents

Electricity and Electric Circuits

- **Atoms, protons, neutrons, and electrons**: All matter (any substance that has mass and takes up space) is made up of tiny particles called atoms. Copper is made up of copper atoms, oxygen is made up of oxygen atoms, and so on. An atom is the smallest particle of a particular substance that can exist. For example, if you split a copper atom into two parts, those two parts are no longer copper atoms, but are something else.

  Each atom has a small central nucleus consisting of sub-atomic particles called protons and neutrons. The nucleus itself is surrounded by even smaller particles called electrons, which whizz around the nucleus in what is known as an electron cloud.

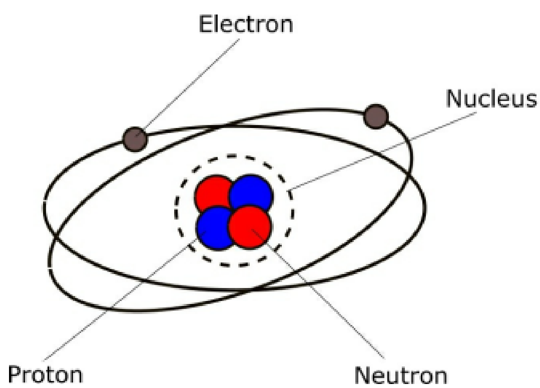


Figure 1 - The structure of an atom

Atoms are really small, ranging in size from about 0.1 to 0.5 nanometers (1×10^{-10} m to 5×10^{-10} m)

That's roughly a million times thinner than a human hair

Protons and neutrons are even smaller, with electrons being the smallest particles

An electron is about 200,000 times smaller than a proton

  Note: A **particle** is either a very small piece of matter.

  Although we call an atom the smallest particle, there are still things smaller. An atom is called the smallest particle because it is the smallest unit of an element that still keeps the chemical identity and properties of that element

  An **atom** is the smallest basic unit of a chemical element that keeps the properties of that element.

- **Nucleus of an atom**: The nucleus of an atom is the tiny, dense core located at the center of the atom. According to the Department of Energy, it holds more than 99.9% of the atom's total mass while taking up a tiny fraction of its space

  The nucleus contains two main types of particles, which are known together as nucleons:

  - Protons
  - Neutrons

  Protons push away from each other because they share the same positive charge. A powerful natural rule called the strong nuclear force glues the protons and neutrons tightly together.

The outside of the nucleus is surrounded by a cloud of tiny, negative particles called electrons.

- **Elements and molecules**: An **element** is a pure substance made of only one type of atom that cannot be broken down into a simpler substance by normal chemical processes

  Elements can exist as collections of billions of individual atoms (like solid copper or gold), or they can bond together into small groups called molecules.

  A **molecule** is a group of two or more atoms held together by chemical bonds

- **Electric Charge (Electromagnetism)**: Protons have a positive (+) electric charge, and electrons have an equivalent negative (-) charge.

  Neutrons are neutral, that is, they don't have an electric charge. Although neutrons are extremely important in chemistry and physics, they are not important in electricity ... so we won't say any more about them. Electrons and protons, on the other hand, form the basis of how electric current flows ... so we will talk more about them.

  Electrons are very, very small ... about 200,000 times smaller than protons. An atom of a particular element normally has the same number of protons as electrons. Copper, which is a very important element in electricity, has 29 protons in its nucleus, and 29 electrons around the outside of the atom.

  There are 118 elements and each one has an atomic number, which is determined by the number of protons in its nucleus. A hydrogen atom has one proton in its nucleus, which means it has the atomic number 1. Helium has two protons in its nucleus and has the atomic number of 2. As mentioned above, copper has 29 protons in its nucleus and has the atomic number 29.

  Protons and electrons have opposing electric charge, which causes them to be attracted to each other. It is this attraction that holds atoms together, with electrons being held in place in orbit around the nucleus.

  The physical interaction of electrically charged particles (protons and electrons) is known as **electromagnetism,** which is one of the fundamental forces of nature.

- **Ions**: Sometimes atoms gain or lose electrons. When this happens, it causes the atom that has lost the electron to not have quite enough electrons, while the atom that has gained an electron now has slightly too many. These atoms are called ions. A positive ion occurs when an atom loses an electron (negative charge), which means that it has more protons (positive charge) than electrons. A negative ion occurs when an atom gains an extra electron so it has more electrons than protons causing it to have a negative charge.

  This process of atoms losing and gaining electrons takes place in a random manner within an element, so some atoms are losing electrons, and some are gaining electrons, causing positive and negative ions to be created all over the place.

- **Cation / Anion**: Ions are divided into two main categories: **cations** (positively charged ions) and **anions** (negatively charged ions).

They can also be categorized as **monatomic** (consisting of a single atom) or **polyatomic** (consisting of multiple atoms bonded together).

- **Conducting and Insulating Elements**: With some elements - copper, for example - the process of electrons randomly jumping around between atoms takes place more easily than with other elements. It is this property of copper that makes it an excellent conductor. Other excellent conductors are silver and aluminium

  Elements that are reluctant to let their electrons move from one atom to another are known as insulators. Materials made of polymers or plastics tend to be very good insulators, which is why they are used as insulation around copper wires. Glass and paper are also excellent insulators.

  There is no such thing as a perfect insulator. This is because even insulators contain small numbers positive and negative ions, which can be used to conduct (carry) an electric current. In addition, all insulators become electrically conductive when a large enough voltage is applied such that electrons are torn away from the atoms. When this happens, it is known as the breakdown voltage of the insulator.

- **Voltage and Current**: Although the random movement of electrons between atoms is all very interesting, it's not a lot of use on its own. However, if we could get them to travel in the same direction, then perhaps we could take advantage of that. In fact, the result of doing this is what creates an electric current within a conductor. What happens is that when all the electrons start travelling in the same direction, they begin to transfer their electromagnetic force through the conductor

  Although each individual electron only travels a few millimetres per second, because this is happening trillions of times simultaneously, the net result is that the flow of electrons takes place at almost the speed of light.

  Getting electrons to all flow in the same direction doesn't happen by accident - we need to do something in order to make it happen. We do this by using voltage.

  A voltage is the difference in electric charge between two points. So, for example, if you have one piece of metal that has an abundance of positive ions (positively charged atoms - they don't have enough electrons), and another piece of metal with an abundance of negative ions (negatively charged atoms - they have too many electrons), a voltage (difference in electric charge) exists between the two pieces. If you then attach a conductor between them, electrons (negatively charged particles) flow from one piece of metal to the other. The flow of electrons continues until all of the atoms in the circuit are electrically neutral.

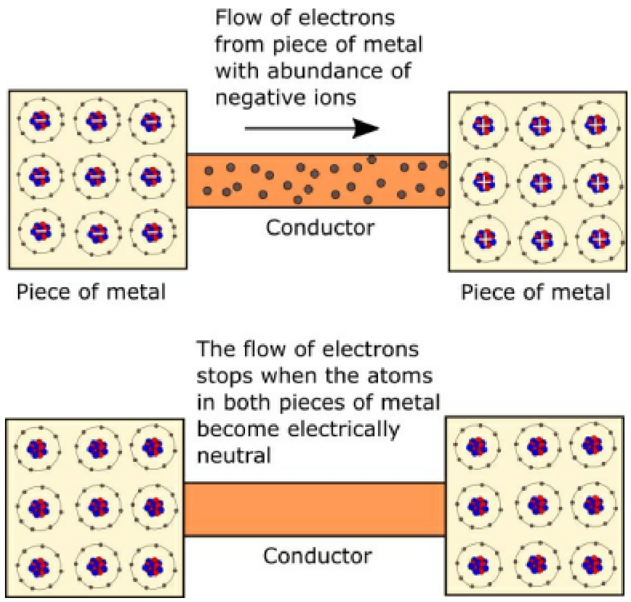


Figure 5 - Flow of electrons between two pieces of metal

Once the electrons stop flowing, they revert to their random movement between atoms as shown earlier.

So, we can conclude that a voltage, which is a difference in electric charge between two points, needs to exist in order for electric current to flow.

Voltage can also be referred to as potential difference. Electromotive force (EMF) is the voltage developed by any source of electrical energy, for example, a battery. It is often defined as the source's electrical potential in a circuit.

- **Measuring Voltage**: Voltage is measured in volts (V). There are a few types of instrument that can be used to measure voltage:

  1. Voltmeter (often integrated into a multimeter)
  2. Potentiometer
  3. Oscilloscope

- **Voltage sources**: Batteries are often used to provide a direct current (DC) voltage for a circuit.

  The voltages supplied by power companies to consumers are much higher than those provided by batteries. The exact voltage supplied to your home or business depends where in the world you live, but is either 110 to 120 volts (alternating current - AC) or 220 to 240 volts (AC).

- **DC (direct current) and ac (alternating current)**: With direct current (DC), the movement of electric charge through a circuit is in one direction only. This is because the voltage remains constant, which keeps the electric charge flowing in the same direction. Batteries are typical sources of direct current.

In an alternating current (AC) circuit, the voltage is periodically reversed, which causes the flow of electric charge to also reverse. This normally happens about 50 or 60 times per second. AC, which is more efficient than DC at transmitting electric charge over long distances, is the form of electric power normally delivered to commercial and residential properties.

As we're dealing with electronic circuits in this book, we only look at DC, battery-driven, circuits.

- **Measuring current**: Current is the flow of electric charge and is measured in amperes (or amps (A)), which is a measure of how many charge carriers (electrons, in general) flow past a certain point in one second. One ampere equals a flow of 6,240,000,000,000,000,000 electrons per second.

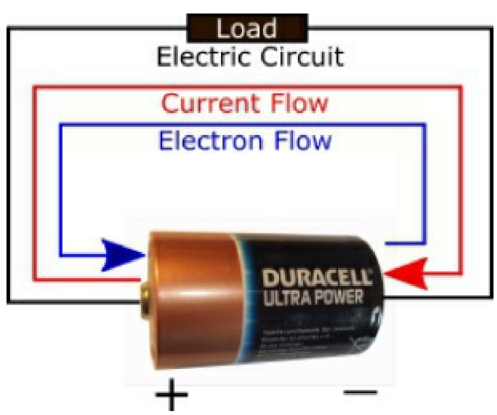
  To measure how much current is flowing through a circuit you can use an ammeter, which is normally integrated into a multimeter

- **Conventional Current Flow**: As we have already established, electrons are negatively charged particles and flow when there is a difference in electric charge (voltage) between two points. The direction of flow is from the negative end of the circuit towards the positive end. This is because opposite electric charges are attracted to each other, so electrons (negative) flow towards the positive charge

  In a battery, for example, electrons flow from the negative terminal, through the circuit, and then back to the positive terminal.

  However, as well as electron flow, there is something called conventional current flow, which flows in the opposite direction to electron flow.

  Conventional current flow is from positive to negative; electron current flow is from negative to positive.



- **More on battery**: A battery uses chemical reactions to create a voltage difference between its terminals

  1. At the negative terminal, a chemical reaction releases electrons.
  2. When a circuit is connected, the battery establishes an electric field through the wire.

3. Because electrons are negatively charged, they move opposite the electric field, from the negative terminal, through the device, to the positive terminal.
4. At the positive terminal, another chemical reaction accepts those electrons

- **Load**: A load is any component that uses electrical energy from the battery to perform some function.

  Never connect the terminals of a battery directly together as this will cause the battery to overheat very quickly and may even cause it to explode. There must always be a 'load' (for example, a light bulb) in between the battery terminals

- **Resistor**: A resistor is a passive two-terminal electronic component that limits or regulates the flow of electrical current in a circuit

  When you are trying to determine where in a circuit to place a resistor, for example, you need it to be in between the battery's positive terminal and the component that the resistor is protecting. This is because the flow of current is coming from the positive terminal.

- **Resistance**: All the components contained in an electric circuit - including the wires contenting those components - resist the flow of current to a certain extent. There is no such thing as a perfect conductor so, even material like copper, offers a certain amount of resistance. Materials that are used as insulators, such as polymers, offer a lot of resistance, which is why they are used as insulators. If you mix a material that is a good conductor with a material that is a good insulator, you end up with something in between the two. This is how resistors work

  If you add up all the resistances of the individual components in a circuit (if they are connected in series), you end up with the overall resistance of the circuit itself.

  The amount of resistance provided by a wire increases as:

  – the length of the wire increases, and
  – the thickness of the wire decreases.

  This means that a long thin copper wire will have a tendency to provide more resistance than a short fat copper wire. However, in the experiments we carry out in this book, the jump wires have a very little resistance and as such their resistance value can be ignored.

  Note: Wires are made of metal, and the atoms inside metals naturally form positive ions because of the way they bond together.

  When electrons flow through a wire they sometimes bump into ions, which make it more difficult for the electrons to flow, causing resistance. In a long wire, this is more likely to happen than in a short wire. Likewise, in a thin wire, there are fewer electrons to carry the current than in a thick wire.

  A thin wire has a smaller cross-sectional area, meaning it has less physical space inside to hold free electrons than a thick wire.

- **Jumping**: To "jump" in electronics means to create a temporary or permanent electrical connection between two points in a circuit using a conductor like a wire or a small plug.

- **Measuring Resistance**: Resistance is measured in ohms. 1 ohm is the amount of resistance required to allow 1 amp of current to flow around a circuit when 1 volt is applied to that circuit.

To measure the resistance in part of a circuit you can use an ohmmeter.

- **Ohm's law**: Ohm's law relates the current, denoted I and the resistance, denoted Ω to the voltage V:

$$V = I\Omega.$$

Thus,

$$I = \frac{V}{\Omega},$$
$$\Omega = \frac{V}{I}.$$

- **Kirchhoff's laws**: Kirchhoff's circuit laws are two fundamental rules used to find currents and voltages in electrical networks.
  - **Kirchhoff's First Law (Current Law / Junction Rule)**: The total current entering a junction or node must equal the total current leaving that junction.

$$\sum I_{\text{in}} = \sum I_{\text{out}} or \sum I = 0.$$

  - **Kirchhoff's Second Law (Voltage Law / Loop Rule)**: The total voltage around any closed circuit loop must equal zero.

$$\sum V = 0.$$

  Sum of voltage rises equals sum of voltage drops

- **Work**: Work is the energy transferred to or from an object when a force causes that object to move through a distance and is measured in **joules** J. One Joule equals one Newton-meter $(N \cdot m)$.

$$W = Fd\cos(\theta).$$

θ is the angle between the force vector and the direction of motion

- **Power**: Power is the amount of work done by an electric circuit and is measured in watts (W).

In this case, electric charges (specifically electrons) are the objects being displaced. A battery can be used as a power source for a small electric/electronic circuit. However, to supply power to homes and businesses, electric generators are used. Electrical power can be carried over long distances and then efficiently converted into other forms of energy such as light, heat or motion.

The amount of power generated by an electric circuit depends on the amount of current passing through the circuit, as well as the amount of voltage used to push the current through the circuit.

$$W = VI.$$

So, for example, a current of 1 A, which is being pushed around a circuit by a 1 V power supply, generates 1 W of power.

In order to do something useful, the electric energy carried by the current needs to be converted into a different form of energy such as heat or light. For example, with an electric heater, a voltage pushes current through heating elements to generate heat; with a light bulb, a voltage pushes current through a filament to generate heat and light. When this happens, the circuit is said to dissipate a certain amount of power in watts. For this reason, heaters and lights are both given power ratings in watts, for example, a 2 kilo-watt (kW) heater, a 40 W light bulb, and so on. The higher the power rating, the more heat or light is generated by the circuit, in other words, the more work is being done.

Power dissipation is the process where electrical energy is converted into heat or other unusable energy and lost from an electrical or electronic system

A **power rating** is the maximum amount of power a device or machine can safely consume, handle, or produce under normal operating conditions

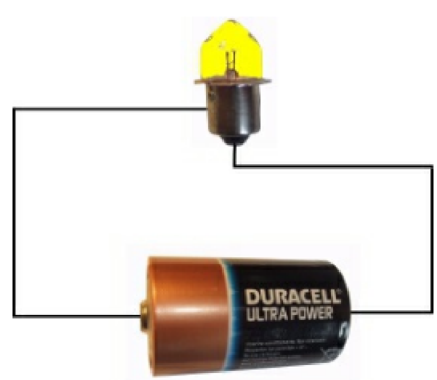
In the same way that you can calculate the power generated by a circuit if you know the voltage and current, you can calculate the current flowing through a circuit if you know the power and voltage. To do this you just need to rearrange the above formula:

$$I = \frac{W}{V},$$
$$V = \frac{W}{I}.$$

Note: One watt is equal to one joule of energy used or transferred per second

$$W = \frac{J}{s}.$$

- **A Basic Electric Circuit**: An electric circuit is a complete loop that starts and ends at the same point and provides a conductive path for an electric current to flow.



In this circuit, current flows from the battery to the light bulb. The current then continues through the bulb and returns to the battery, completing the circuit.

A light bulb has two electrical connectors (terminals) on it: one is at the end of the base and is normally a round dot, and the other is on the side of the base, which is sometimes a thread. It doesn't matter whether you connect the battery's positive terminal to the side of the light bulb or the bottom - the circuit will work either way.

Things to note:

- **Voltage** - The battery supplies a voltage that causes current (electrons) to flow around the circuit. The electrons leave the negative terminal of the battery and travel around the circuit (through the light bulb) until they reach the positive terminal of the battery.
- **Closed Circuit** - The circuit is closed - it forms a complete loop, starting and ending at the same place (the battery). If there was a break (or gap) in the circuit it would not work.
- **Load** - In order to do something useful, an electric circuit needs to have a load. In our case, the load is the light bulb. In electronic circuits, the load is made up of components such as resistors, capacitors, light emitting diodes (LEDs), buzzers, motors, and light sensors.

• **Adding a Switch**: In its current form the circuit is not particularly practical because there is no way of turning off the light bulb without disconnecting the battery. To address this short-coming we can add a switch

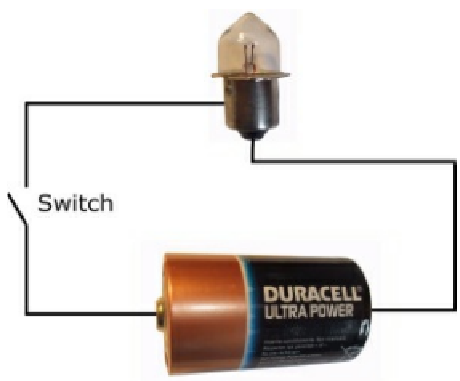


Figure 9 - Adding a switch to the circuit

What we have done by introducing the switch is to create a situation whereby we can turn the closed circuit into an open circuit. This prevents the current from flowing through the light bulb, which turns it off.

• **Short Circuit**: Electric circuits can become very complicated and ensuring that the current flows around those circuits correctly is a tricky business and sometimes short circuits occur. This is when the current takes an alternative route to the one you want it to, and in doing so incorrectly bypasses some or all of the components that make up the system. The circuit we have been looking at so far is very simple, and introducing a short circuit is very easy to do, as shown below.

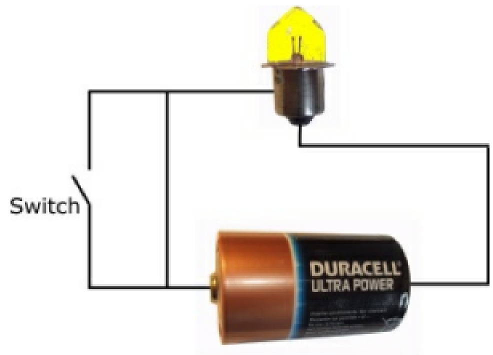
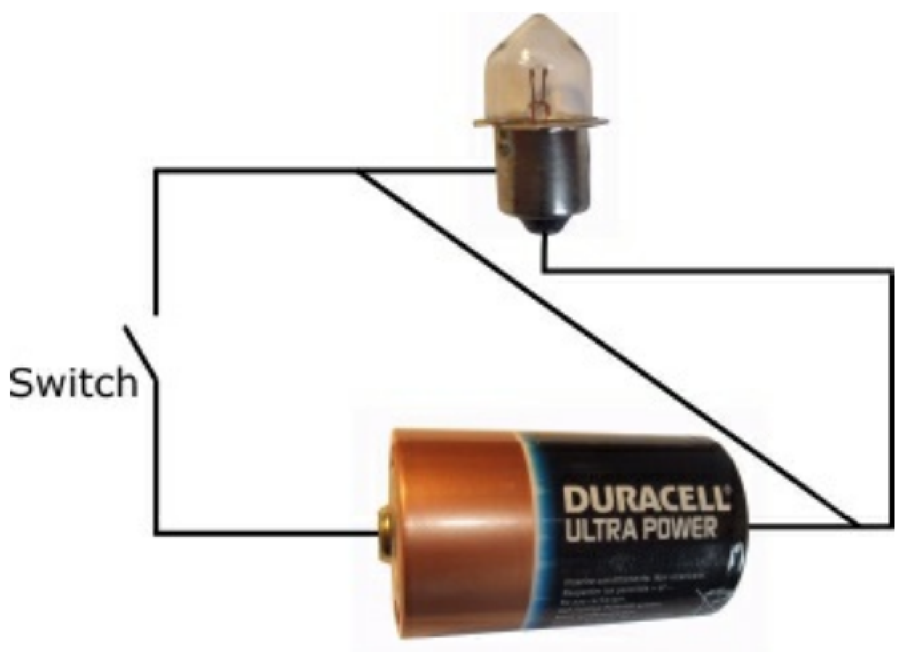


Figure 10 - A short circuit

The short circuit shown in Figure 10 causes the switch to be bypassed, meaning that whether it is open or closed, the light bulb will always be on. Although this short circuit would be annoying if you had designed the circuit, it would not cause any damage to the light bulb.

When the switch is closed, the current will flow down both paths, which then converge on their way to the light bulb.

Now, consider



The short circuit shown above is more serious because it bypasses the light bulb (the circuit's load). When the switch is closed, the circuit will get very hot as the current races between the two terminals of the battery. Some current will flow through the light bulb when the switch is closed, but most of it won't.

- **Why do things get hot**: A battery has a very small amount of internal resistance. We can use Ohm's law to calculate how much current will flow around the circuit when the two battery terminals are connected to each other. (I'm assuming that the resistance of the wire is negligible so I haven't included it in the calculation.). If volts = $1.5V$, resistance = $140m\Omega$,

  $$I = 1.5/0.140 = 10.71 \text{ amps.}$$

  This is a lot of current to pass through the battery, which will cause a very rapid build up of heat in the circuit and may even cause the battery to explode.

- **The Battery**: As with all of the circuits we describe in this book, the voltage source (the power source) is provided by a battery. The battery shown in the above circuits is a 1.5 volt 'D'. If we add a second battery in series with the first one, the light bulb will glow more brightly because we have increased the amount of voltage pushing the current around the circuit.

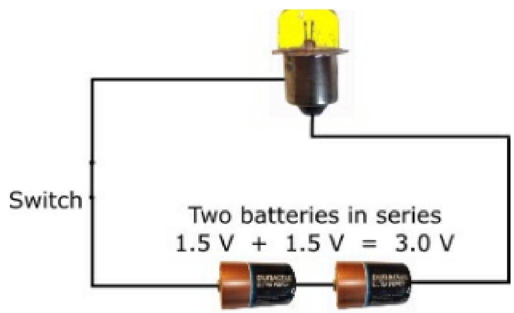


Figure 12 - Two Batteries in Series

  On the other hand, if we add more light bulbs to the circuit, they will become dimmer because the voltage is having to push the current through a bigger load.

- **Electronic circuits**: Electronic circuits are made up of various components that work together to create a circuit that does something. Exactly what that "something" is depends on the circuit. Electronic components can be categorized into two groups:

  - Discrete components
  - Integrated circuits (ICs)

  Discrete components include items such as resistors, capacitors, and diodes. Integrated circuits (ICs) are tiny circuits etched onto silicon and are often referred to as chips. They are normally used to perform a particular function, for example a timer, and can be combined with discrete components to form larger circuits.

  ICs have pins on them that provide access to the various parts of the IC's internal circuit. Getting the IC to function properly requires correct use of these pins.

- **Schematic Circuit Diagrams**: Although the circuit diagrams we have looked at so far in this chapter look quite nice in that you can easily recognize the battery and the light bulb, once a circuit starts to become more complicated, this approach would soon become quite cumbersome. So, instead, schematic circuit diagrams are normally used in which components are represented by symbols, with lines between them to show how they are connected

The idea is that the symbols convey enough information to enable someone to understand and build the circuit, but do not contain any unnecessary detail regarding the components themselves.

The lines that join the symbols represent wires on a breadboard or traces of copper on a printed circuit board.

Circuit traces are the thin, conductive copper pathways on a printed circuit board (PCB) that act as wires to connect electronic components

A couple of things to note about schematic circuit diagrams:

- They always depict conventional current flow (positive to negative) rather than electron current flow (negative to positive).
- The layout of components in a schematic circuit diagram does not have to match the physical layout of the components when the circuit is built.

Circuit schematics depict conventional flow because historical convention established current as moving from positive to negative before electrons were discovered

- **Coulomb and electric charge**: Electric charge is a basic physical property of matter that causes objects to feel a force when near other charged matter. Charge is either positive or negative. Protons carry a positive charge and electrons carry a negative charge.

  Like charges push away from each other (repel), and opposite charges pull toward each other (attract).

  Charge is denoted Q and measured in **coulomb** C, with

  $$C = A \cdot s.$$

  Thus, one amp is one coulomb per second, and:

  $$Q = It.$$

  Consider a current of 3 A that flows for 2 s

  $$Q = 3A \cdot 2s = 6A \cdot s = 6\ C.$$

  Current is therefore the rate at which charge flows

  $$I = \frac{Q}{t}.$$

- **Energy**: Energy is a fundamental quantitative property that describes a system's capacity to perform work or cause change, measured in joules J.

  - **Kinetic Energy:** The energy of motion.
  - **Potential Energy:** Stored energy based on an object's position, state, or configuration.

  $$J = \frac{kg \cdot m^2}{s^2}.$$

- **Electrical potential**: Electrical potential describes how much electrical potential energy a unit of positive charge would have at a particular location.

$$\text{Electrical potential} = V = \frac{\text{potential energy}}{\text{charge}} = \frac{U}{q}.$$

  Its unit is the volt

$$V = \frac{J}{C}.$$

- **More on voltage**: Electrical potential applies to a location. Voltage is the difference in potential between two locations:

$$\Delta V = V_B - V_A.$$

  If point A is at 0V and point B is at 5V, then B is 5V higher in electrical potential than A. The voltage between them is 5V.

  Potential must be measured relative to a reference. In circuits, we normally choose one point as ground and assign it a potential of 0 V.

  A positive charge naturally moves from higher electrical potential toward lower electrical potential. Electrons are negatively charged, so they tend to move in the opposite direction, from lower potential toward higher potential.

- **Parallel vs series circuits**: A parallel circuit is an electrical circuit where two or more components are connected across the same two nodes, creating multiple paths for the electric current to flow.

  - **Same Voltage (V):** The voltage across every single parallel component is exactly the same. If a 12V battery is connected to three parallel resistors, each resistor experiences the full 12V.
  - **Split Current (I):** The total current entering the parallel section splits up into the different branches. Paths with less resistance get more current, and paths with more resistance get less current.
  - **Independent Operation:** Because each branch has its own direct path to the power source, if one component breaks or is turned off, the other branches keep working perfectly. This is how the electrical wiring in your home is designed.

  In a series circuit, all components are connected end-to-end in a single line, forming a single path for the electrical current to flow.

  - **Identical Current (I):** The current flowing through every single series component is exactly the same ($I_{total} = I_1 = I_2 = I_3$). There are no alternate routes or branches, so current cannot split up.
  - **Shared Voltage (V):** The total voltage from the power source is split across the components ($V_{total} = V_1 + V_2 + V_3$). Components with higher resistance will take a larger share of the voltage.
  - **Cumulative Resistance (R):** The total resistance of the circuit is simply the sum of all individual resistances ($R_{total} = R_1 + R_2 + R_3$). Adding more resistors in series increases the total resistance and decreases the overall current.Dependent Operation: Because there is only one path, if any single component burns out, breaks, or is disconnected, the entire circuit is broken and all components stop working. (This is why one broken bulb on an old string of Christmas lights turns off the whole strand).

  - **Dependent Operation:** Because there is only one path, if any single component burns out, breaks, or is disconnected, the entire circuit is broken and all components stop working. (This is why one broken bulb on an old string of Christmas lights turns off the whole strand).

- **Combination circuit**: When you have a circuit with both series and parallel components, it is called a combination circuit (or series-parallel circuit).

  To solve these, you work backward like a puzzle, simplifying the parallel parts first until you are left with just one simple series chain.

  - **Group the parallel parts**: Identify which resistors are parallel to each other
  - **Find the Group Voltage:** Remember that parallel components share the exact same voltage.
  - **Add the series parts:** Now treat that parallel group as one single block. If that block has 5V and it is in series with resistor R_3 (which has, say, 3V), you add them together:

  $$\text{Total Voltage} = \text{Parallel Group Voltage} + \text{Series Component Voltage}.$$

- **Combining parallel resistors**: Adding resistors in parallel always decreases the total resistance of the circuit because it provides multiple pathways for the electrical current to flow

  To find the total equivalent resistance R_{total}) of resistors in parallel, you add their inverse (reciprocal) values together and then invert the final sum:

  $$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots + \frac{1}{R_n}.$$

  Where R_1, R_2, etc., are the resistance values of the individual resistors.

  Note: The current changes when parallel resistors are combined.

Equipment

- **Intro**: To carry out electronics experiments you need to have a variety of electronic components and equipment available to you. The more components you have in your set, the more you will be able to play around with circuits, but initially you'll be able to get by with a fairly modest set.

  The list of components and equipment in the following table is reasonably extensive, so don't worry about getting everything straight away. In fact, it's probably better to acquire stuff slowly rather than going out and buying everything in one go. That way, you'll be able to spend more time getting components and equipment that are right for you.

  - **Breadboards**: A breadboard enables you to prototype electronic circuits without the need to solder components in place. With a breadboard you can insert a component into position and then remove it and put it somewhere else.

    Beneath the holes are a series of internally connected rows and columns that enable components to be connected to each other by jumper wires.

    To supply power to the breadboard you normally use a battery.
  - **Battery**: All of the electronic circuits described in this book use a battery (or batteries) as the power source.
  - **Battery clip**: A breadboard battery clip (or battery snap/holder with pins) is a connector that attaches to a battery and features rigid male pins designed to plug directly into a solderless electronic breadboard
  - **Battery box**: A breadboard battery box is a specialized battery holder with wire leads that feature solid male pins on the ends, allowing it to plug directly into a solderless electronic breadboard without any stripping or soldering.
  - **Resistors**: Pretty much every electronics circuit uses resistors, which are small devices used to control the flow of current in a circuit.

    An electronic circuit is made up of components that require a certain amount of current in order to operate correctly. Resistors enable you to control how much current flows into components to ensure that they work as intended and do not get damaged.

    Resistors are very cheap and when you start building circuits you will notice that you very quickly accumulate a large selection of resistors.

    You should also get yourself a few variable resistors (potentiometers). With this type of resistor you can manually adjust the resistance value from a minimum to a maximum value

    With an LDR (light dependent resistor), the amount of resistance provided by the resistor varies with the amount of light detected.

    You could also consider getting a few thermistors. A thermistor is a resistor in which the amount of resistance provided depends on the temperature.
  - **Diodes**: A diode is an electronic component that lets current flow in one direction only

A diode has two leads (or pins): the anode and the cathode. You can normally identify which is which because there will be some kind of marker on the diode such as a silver or colored band, which is next to the cathode.

If the anode is connected to a higher voltage than the cathode, current flows through the component (from the anode to the cathode). If the cathode is connected to a higher voltage than the anode, current does not flow through the component.

- **Light emitting diodes (LED)**: As with normal diodes (mentioned above), light emitting diodes (LEDs) have an anode and cathode and only allow current to flow in one direction through the component. However, different to normal diodes, LEDs emit light when current passes through them.
- **Capacitors**: Capacitors are used to store electrical energy and are very widely used in electrical and electronic systems.

  A polarized capacitor has a positive pole and a negative pole and has to be inserted the correct way round in a circuit in order to work.
- **Transistors**: Transistors are the building block of modern electronic devices and are used to switch or amplify electronic signals.
- **Push buttons / switches**: A push button is a switch mechanism that lets you open or close a circuit.
- **Jump wire**: A jump wire is an electrical wire with a connector at each end. Sometimes the wires can be grouped together within a cable, but in this book we will just use individual jump wires.

  The connectors can be inserted into the holes on a breadboard to connect components and create a circuit.

  Different colored wires are available. The colors can be used to indicate different signals or paths if required, but as the wires are the same thickness, they are all the same from an electrical point of view. Different lengths of wire are also available.
- **Crocodile (or alligator) clip**: A crocodile (or alligator) clip is a sprung metal clip with long, serrated jaws which is used for creating a temporary electrical connection. It gets its name from the resemblance of its jaws to that of a crocodile's.

  Crocodile clips are very useful when you're using a multimeter to measure current, voltage, or resistance in a circuit because sometimes you feel as though you need an extra pair of hands when you're trying to take measurements.
- **Small DC motor**: Direct Current (DC) motors are good fun to play with so you should certainly get at least one as part of your electronics laboratory.

  Ideally, look for something that operates in the 6 to 15 V range so it is suitable for use with a 9 V battery.

  You should also get a fan blade that you can attach to the motor's drive shaft. This not only enables you to see when the motor is running, but it also lets you generate electricity using wind power
- **Terminal block**: A screw terminal block is a type of electrical connector where a wire is held by the tightening of a screw.

- **Multimeter**: A multimeter is a piece of test equipment that enables you to check the voltage (in volts), current (in amps) and resistance (in ohms) in a circuit. It is in fact a combination of three separate test tools
    - an ammeter, which measures current,
    - a voltmeter, which measures voltage, and
    - an ohmmeter, which measure resistance.
- **Oscilloscope**: An oscilloscope is a type of electronic test instrument that allows observation of varying signal voltages, usually as a twodimensional plot of one or more signals as a function of time.

- **Guidelines**:
  - Always connect up all the components in your circuit before you supply any power to it.
  - Once power is being supplied, touch the battery and other components to check that nothing is getting hot. Do this on a regular basis while you are working on the circuit.
  - When you have finished working on the circuit, disconnect the battery from the circuit.
  - If you have battery clip connected to the battery terminals, remove it when the battery is not supplying power to a circuit. If the clip's two wires touch each other while the clip is connected to the battery, this will cause a short circuit and the battery will get very hot, very quickly.
  - Disconnect the power source straight away if you smell burning.
  - Try to avoid working in your electronics lab when there is no-one else in the house.
- **Hidden high voltages**: Depending on the circuits you build you may well create situations whereby hidden high voltages exist in the circuit. Pay special attention to any circuits that contain capacitors - especially big ones - as these can store electrical energy long after the battery power supply has been removed from the circuit.

Resistors

- **Intro**: Resistors are small devices used to control the flow of current in a circuit, and are used in pretty much every electronics circuit. They are made from a combination of material that conducts current easily (for example, a conductor such as carbon) and material that does not conduct current easily (for example, an insulator such as ceramic). The mixture of these two materials determines the amount of resistance provided by the resistor. A mixture high in ceramic produces a resistor that imposes a high resistance on the flow of current, while a mixture high in carbon produces a resistor that lets current flow relatively easily.
- **Why Resistors are Important**: An electronic circuit is made up of components that require a certain amount of current in order to operate correctly. If the amount of current is too low, the component may not work properly or perhaps not at all. For example, if a light emitting diode (LED) requires 25 milliamps of current in order to illuminate properly, but the circuit is only supplying 10 milliamps, the LED will not be very bright. On the other, if you allow too much current to flow through a component, you run the risk of destroying that component. For an LED it might mean that it shines very brightly and doesn't last very long, or it might even cause the LED to go bang.

  By using Ohm's Law you can calculate the size of resistor you require in order to allow the correct amount of current to flow through a component.
- **Resistor Sizes**: In 1952 the IEC (International Electrotechnical Commission) defined a standard for resistance values and tolerances to ease the mass production of resistors. These are referred to as preferred values or E-series, and are published in the IEC 60063:1963 standard.

  The following E series exist

  - E6 series (tolerance 20%)
  - E12 series (tolerance 10%)
  - E24 series (tolerance 5% and 1%)
  - E48 series (tolerance 2%)
  - E96 series (tolerance 1%)
  - E192 series (tolerance 0.5%, 0.25% and 0.1%)

  Don't worry about what resistor tolerance means at this stage, we cover that later on.

  The following table shows the resistor values that conform to the E12 series, which is the most popular of the E series.

1.0	10	100	1.0 k	10 k	100 k	1.0 M
1.2	12	120	1.2 k	12 k	120 k	1.2 M
1.5	15	150	1.5 k	15 k	150 k	1.5 M
1.8	18	180	1.8 k	18 k	180 k	1.8 M
2.2	22	220	2.2 k	22 k	220 k	2.2 M
2.7	27	270	2.7 k	27 k	270 k	2.7 M
3.3	33	330	3.3 k	33 k	330 k	3.3 M
3.9	39	390	3.9 k	39 k	390 k	3.9 M
4.7	47	470	4.7 k	47 k	470 k	4.7 M
5.6	56	560	5.6 k	56 k	560 k	5.6 M
6.8	68	680	6.8 k	68 k	680 k	6.8 M
8.2	82	820	8.2 k	82 k	820 k	8.2 M

Table 2 - E12 Resistor values

In the above table 'k' stands for kilo-ohm (1,000 ohms), and 'M' stands for mega-ohm (1,000,000 ohms).

- **Resistor Color Coding**: A resistor has a series of colored bands around it that indicate the amount of resistance provided by the resistor (the value of the resistor), as well as its tolerance, which is a measure of its accuracy - in other words, how close to the indicated resistance value the resistor actually is.

we can see that it has four bands:

  - Yellow
  - Violet
  - Yellow
  - Silver

The majority of resistors have four bands; the first three indicate the resistance value and the fourth indicates the tolerance. If a resistor has five bands, the first four represent the value and the fifth indicates the tolerance.

It can sometimes be tricky to determine from which side you should read the colors. The way to do it is to try to determine the band that is closest to the end of the resistor and start with that color. If the resistor has a gold or silver band on it

then you should position that band on the right-hand side when you are working out the resistor's value. This is because resistor values never start with a gold or silver band, so those colors can never be on the left-hand side.

There are plenty of online calculators to help you work out resistor values and colours - whether you have a resistor and you're not sure what its value is, or if you know what value you want but you're not sure what the resistor looks like. You can also use a multimeter to quickly measure the resistance of a resistor.

Color	Digit 1	Digit 2	Digit 3*	Multiplier	Tolerance
Black	0	0	0	x 10^0	
Brown	1	1	1	x 10^1	± 1% (F)
Red	2	2	2	x 10^2	± 2% (G)
Orange	3	3	3	x 10^3	
Yellow	4	4	4	x 10^4	
Green	5	5	5	x 10^5	± 0.5% (D)
Blue	6	6	6	x 10^6	± 0.25% (C)
Violet	7	7	7	x 10^7	± 0.1% (B)
Grey	8	8	8	x 10^8	± 0.05% (A)
White	9	9	9	x 10^9	
Gold				x 0.1	± 5% (J)
Silver				x 0.01	± 10% (K)
None					± 20% (M)

Table 3 - Standard Resistor Color Code Table

* This digit only applies to 5-band resistors.

- **Resistor Tolerance**: We have established from the previous sections that as well as having a resistance value in ohms, resistors also have a tolerance rating. What this means is if you use a 100 Ω resistor that has a 10% tolerance, the actual resistance value will be somewhere between 90Ω and 110Ω.

  In many electronics circuits a tolerance of 5% or 10% is acceptable, and that is certainly the case with all the circuits we use in this book. If, when you design and build your own circuits, you need a higher degree of accuracy (in other words, a smaller tolerance), you can simply use an E series resistor that matches your requirements

  Note that you can get resistors outside of these standard series, for example, you can buy E12 series resistors that have a tolerance of 5%.

- **Resistor Power Rating**: One more thing to consider when using resistors is their power rating, which is measured in watts (W) and will typically be 1/8 (or 0.125) W or 1/4 (or 0.25) W.

The way a resistor works is to restrict the flow of current that passes through it, which causes the resistor to heat up. If the resistor heats up too much it will burn up. The amount of power a resistor can handle before it burns up is calculated using the following formula:

$$W = VI.$$

Assume that a 470Ω is being used in a circuit, and it has $9V$ across it. Then,

$$9\ V = I(470\ \Omega) \implies I = \frac{9}{470}\frac{v}{\Omega} \approx 0.019\ A.$$

Thus, the current is roughly 0.019 amps. From this,

$$W = 9\ V \cdot \frac{9}{470}\ A = \frac{81}{470}\ \frac{V}{A} \approx 0.172\ W.$$

So, the power dissipated by the resistor is 0.172 watts.

For a resistor with a power rating of 1/8 (0.125) W, this would be too much power to dissipate. However, for a 1/4 (0.25) W resistor, this would be acceptable.

Exceeding a resistor's power rating causes it to overheat, permanently change its resistance value, burn out, or potentially catch fire

- **Identifying the Power Rating**: When you buy a pack of resistors, the power rating will be stated on the packaging. However, if you have a box of assorted resistors just lying around, you can't tell what the power rating of a resistor is by looking at it - unlike the resistance value and tolerance rating, there is no indication on the resistor as to its power rating. You can, however, get an idea of a resistor's power rating based on how big the resistor is - bigger resistors have bigger power ratings.
- **Potentiometers**: All of the resistors we have looked at so far have had a fixed resistance. It is possible though to have a resistor that has a variable resistance; such resistors are called potentiometers (or pots, for short). Typical uses of a potentiometer are:
  - volume control on a radio
  - dimmer switch for a light
  - thermostat

  A potentiometer can be used in two ways in a circuit:
  - as a variable resistor (rheostat), or
  - as a voltage divider.

  A rheostat is a two-terminal variable resistor used to control the flow of electric current in a circuit by manually changing its resistance

  We will just concentrate on how potentiometers are used as variable.
- **Pins**: Pins on an electrical component are small metal contact points that create physical and electrical paths for power, signals, or data to flow between the component and a circuit board. They carry electrical current, voltage, or digital data signals into and out of devices like microchips, resistors, and connectors.

Voltage is always a difference between two points. You cannot measure the voltage of just one single spot without a baseline.

Ground acts as the ”zero line” or 0V. If ground is 0V and another part of the circuit reads 12V, you know there is a 12-volt push between those two spots.

- **Ground (GND)**: In electronics, ground (often labeled as GND) is the zero-volt reference point used to measure all other voltages in a circuit and provides a common return path for electric current
- **Using a Potentiometer as a Variable Resistor (Rheostat)**: To use the potentiometer as a variable resistor, only two pins are used: one of the outside ones and the center pin. The position of the wiper determines how much resistance the variable resistor provides.

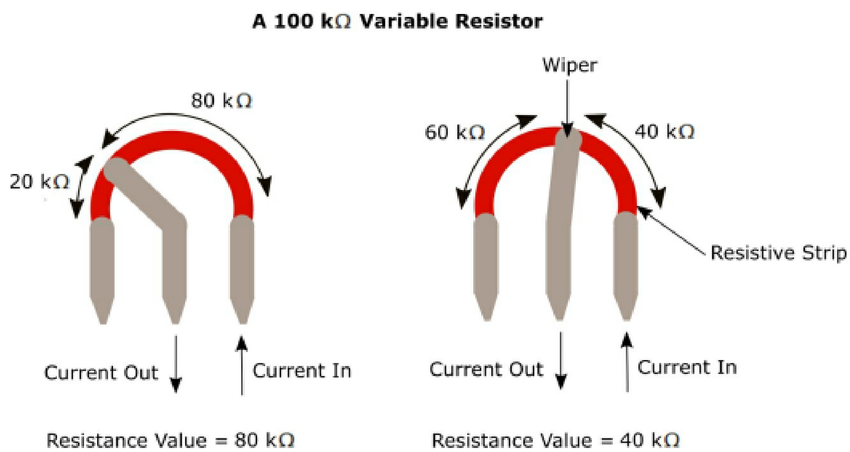


Figure 15 - A 100 k Ω variable resistor

A 100 $k\Omega$ variable resistor, as shown in Figure 15, has a maximum resistance of 100 $k\Omega$ and a minimum of 0Ω. This means that by changing the position of the wiper, the resistance provided is somewhere between 0Ω and 100 $k\Omega$.

To adjust the resistance of a variable resistor, you normally turn (rotate) a knob, but with some variable resistors you need to move a slider.

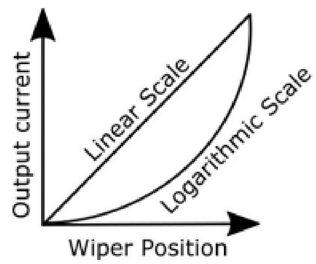
- **Variable Resistor Ratings**: As with normal ’fixed-value’ resistors, a variable resistor has a resistance value, for example, 100 $k\Omega$. The way this works is that the resistance value between one of the fixed terminals and the wiper terminal, plus the resistance value between the other fixed terminal and the wiper terminal, always add up to the total resistance value of the resistor.

Figure 15 assumes that the scale between 0 and 100 $k\Omega$ is a linear one. In other words, if the wiper is 25% of the way round the resistive strip, the resistance value to the left of the wiper terminal is 25 $k\Omega$ and the value to the right is 75 $k\Omega$; and if the wiper is half round the strip, the two resistance values are both 50 $k\Omega$.

It is also possible to have a situation whereby a logarithmic (taper) scale is used rather than a linear one

When a logarithmic scale is used, the output current initially increases very slowly as the wiper position is adjusted, but increases at a high rate towards the end of the wiper adjustment.

Logarithmic scales are typically used to control the volume on audio devices.



- **Photoresistor / Light Dependent Resistor (LDR)**: As the name suggests, the amount of resistance provided by an LDR depends on how much light is detected by the LDR. In low light conditions, the resistance of an LDR is very high, but when a torch is shone on it, for example, the resistance drops dramatically to let current flow through the device.
- **Thermistor**: A thermistor is a type of resistor in which the amount of resistance provided by the device varies according to the surrounding temperature.
  - At low temperatures, the resistance of a thermistor is high, and only a small amount of current can flow through it.
  - At high temperatures, the resistance is low, allowing more current to flow through the device.
- **Schematic Diagram Symbols**:

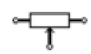
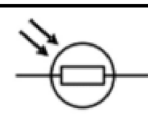
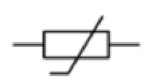
Component	Symbol	Identifier
Fixed Value Resistor (IEEE)		R, R1, R2, R3, and so on
Fixed Value Resistor (IEC)		
Variable Resistor / Potentiometer (IEEE) 3 Terminal		
Variable Resistor / Potentiometer (IEC) 3 Terminal		
Variable Resistor / Rheostat (IEEE) 2 Terminal		
Variable Resistor / Rheostat (IEC) 2 Terminal		
Photoresistor / Light Dependent Resistor (LDR) (IEC)		
Photoresistor / Light Dependent Resistor (LDR) (IEEE)		
Thermistor		

Table 6 – Resistor schematic symbols

Where there is a choice, in this book we use the IEEE symbols.

- **Schematic diagram symbols for power supply**:
  - **DC voltage source:** A circle with + and − signs. The voltage value is usually written beside it, such as 12 V.
  - **Battery:** Alternating long and short parallel lines. The long line is positive; the short line is negative.
  - **Power rail:** An upward-pointing symbol or labeled connection such as VCC, VDD, +5V, or +12V.
  - **Ground/return:** Usually three horizontal lines that become shorter toward the bottom. It represents the circuit's reference voltage, normally 0 V.

In a battery symbol, each pair of parallel lines represents one electrochemical cell:

- The long line represents the cell's positive terminal (+).
- The short line represents the cell's negative terminal (−).

A single long–short pair represents one cell. Several alternating long and short lines represent multiple cells connected in series, which is technically a battery.

Capacitors

- **Intro**: A capacitor is a discrete component that can store an electric charge. Capacitors are available in different sizes, with bigger capacitors able to store more charge than smaller ones. Capacitors are also sometimes known as condensers.

  Capacitors are used in lots of electronic equipment, including televisions, radios and cameras. When your camera uses a flash, it's a capacitor that is used to store the electric charge which enables the flash to happen.

  A capacitor is similar to a battery in that it is able to store electrical energy. However, in a battery the energy is generally released quite slowing, perhaps over a period of days, weeks or even years. Capacitors on the other hand tend to release their energy very quickly - normally within a few seconds or less.

  Capacitors get their electrical energy from another energy source in an electric circuit, such as a battery in the case of a digital camera. Adding electrical energy to a capacitor, which can take a few seconds, is known as charging; releasing that energy, which happens very quickly, is known as discharging.

- **Charging and Discharging a Capacitor**: A capacitor consists of two conductors that allow electric current to flow, separated by an insulator that does not easily allow electric current to flow. The two conductors are known as **plates**, and the insulator is called the **dielectric**.

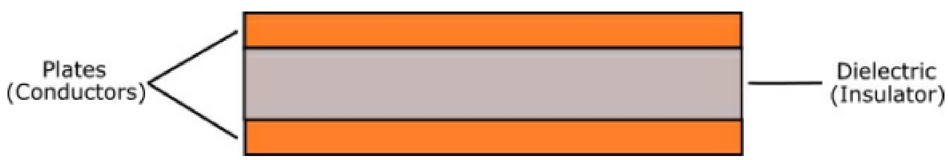


Figure 19 - The parts of a capacitor

  To charge a capacitor you connect it to an electric circuit. When power is applied to the circuit, an electric charge gradually builds up on the plates. One plate gains a positive charge and the other gains an equal but opposite (negative) charge. When the power is disconnected from the circuit, the capacitor holds (keeps) its charge. If the charged capacitor is then connected to a device such as a flash bulb in a camera, the charge flows quickly from the capacitor until there is none remaining on the plates.

  The insulator (called a dielectric) in a capacitor stops direct electric current from flowing between the two conductive plates, which allows the device to store electrical energy. It prevents electrons from jumping straight across from one plate to the other. This forces charges to build up on the surfaces of the plates.

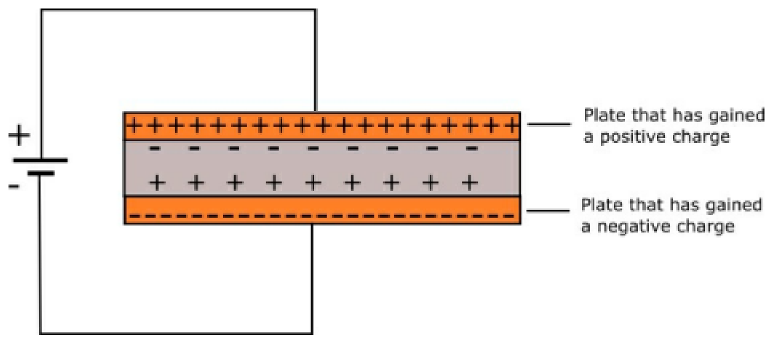


Figure 20 - Charging a capacitor

The positively charged plate and the negatively charged plate are attracted to each other. Separating them requires work to be done against that attraction. The energy required to do this is stored in the electric charges in the two plates as potential energy. This is the energy that is stored when the capacitor is charged, and then released when the capacitor is discharged.

Three things determine how much electric charge a capacitor can store:

- The size of the plates - the bigger the better.
- How close the plates are together - the closer the better.
- How good the insulator (the dielectric) is. Examples are: air, paper, glass, and rubber.

- **Note: Free electrons**: Free electrons are electrons that are not bound to any single atom or molecule and can move around easily

- **Conservation of charge**: The total electric charge in an isolated system never changes, meaning charge can neither be created nor destroyed, only transferred from one object or system to another. The total sum of positive and negative charges before an interaction always equals the sum after the interaction.

  Whenever a charged particle is created or destroyed, an equal and opposite charge appears or disappears at the same time, keeping the net change at zero

  $$Q_{\text{initial}} = Q_{\text{final}}.$$

  If a process creates a positive charge, it must simultaneously create an equal negative charge to balance it out.

- **Electrostatic induction**: Electrostatic induction is the redistribution of electric charge in an object caused by the electric field of a nearby charged object.

- **Note about negative terminals**: The negative terminal is the point in a power source or circuit where electrons build up and flow out to power electrical devices. It supplies a surplus of negative charges (electrons) that travel through a complete circuit.

  The negative terminal is the specific contact point on a battery or power supply where the voltage is lowest.

- **Negative terminals vs ground**: The negative terminal is the source connection where electrons exit a DC circuit, while ground is a chosen reference point designated as zero volts

- **A Bit More About How Capacitors Work**: As we know, electrons have a negative charge and protons

  Opposite charged particles are attracted to each other - electrons like protons, and protons like electrons - but particles with the same charge are repelled by each other - electrons repel electrons, and protons repel protons. This attraction and repulsion is known as electromagnetic force, and it causes all particles to be surrounded by an electric field.

  Electric current needs a conductor in order to flow. An electric field, on the other hand, does not. This phenomenon is what capacitors use, whereby the electric field created by the particles in one plate extends across the dielectric and causes particles in the second plate to be attracted or repelled.

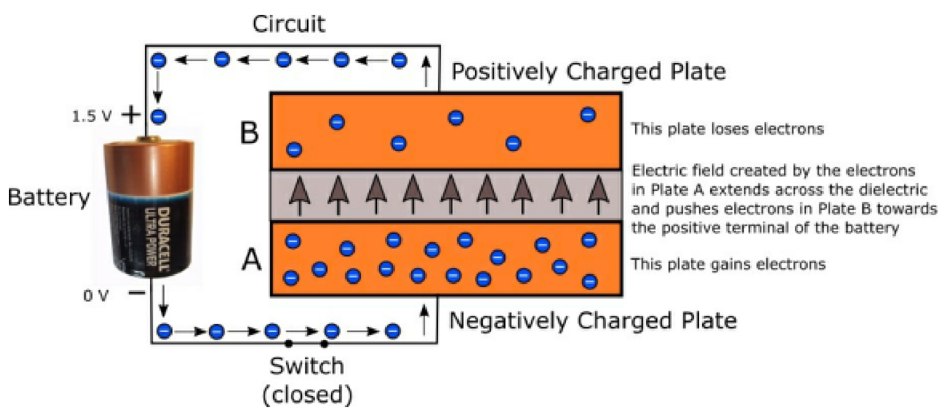


Figure 21 - Charging a capacitor

we can see that applying a voltage across the two plates of the capacitor causes electrons to move from one plate to the other. Plate B (which loses electrons) becomes positively charged, while Plate A (which gains electrons) becomes negatively charged. This creates a voltage between the two plates, which increases until the voltage across the two plates is the same as the voltage across the two terminals of the battery - 1.5 V in our case. At this point the capacitor is charged.

A capacitor stops getting charge when the voltage across its plates becomes equal to the voltage of the power source

If the power supply (the battery) is now removed from the circuit - by opening a switch, for example - the capacitor remains charged and can be used to supply a quick burst of current as and when required in another part of the circuit. When this happens, the capacitor becomes discharged and the plates return to their original state whereby there is no voltage across the two plates, and no current flows.

- **Electric field**: The electric field that exists between the particles in one plate and those in the other plate stores potential energy. It is the electric field that creates the voltage across the two plates. As the capacitor discharges and the voltage across the plates is reduced, the electric field disappears. By this stage, the energy that the electric field stored will have been converted into other forms of energy such as heat and light.

  A Charged Particle creates an electric field around itself. If you bring another charge near it, the field will either push it away or pull it in.

  In a capacitor, you have a plate covered in positive charges directly facing a plate covered in negative charges.

  Because they are trapped on separate plates, they cannot touch, but they still pull on each other.

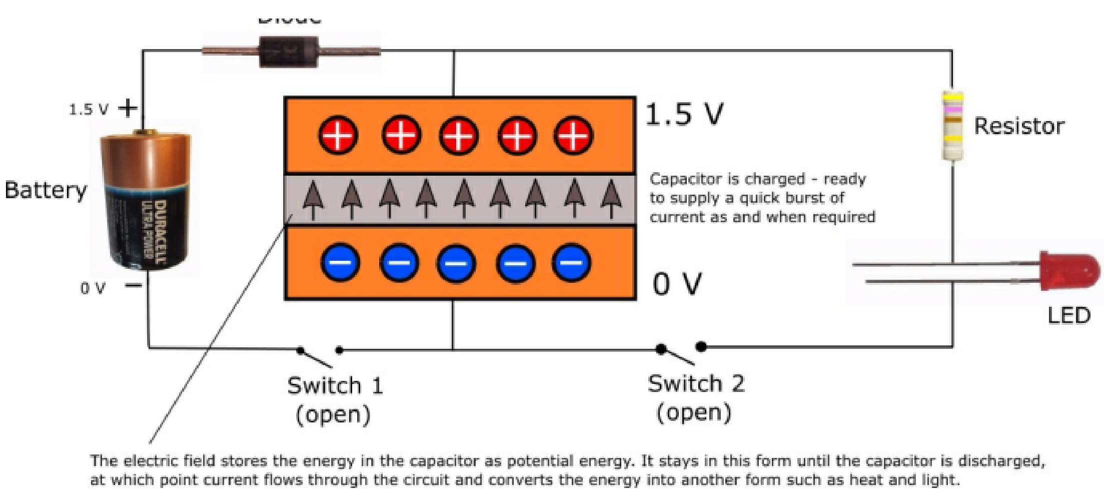
  This intense pull across the empty gap creates a concentrated electric field.That field holds energy in place, like a stretched rubber band, until you give the charges a path to flow.

  Because they are pulling hard but cannot move, they build up **potential energy**.

  The moment you connect the capacitor to a circuit (like a wire leading to a lightbulb), you provide a ”detour.”

  The electrons on the negative plate realize they can take the wire route to get to the positive plate.They rush through the wire to meet the positive charges.

- **Capacitor example**:



The resistor is included in the circuit to stop the LED from burning out when the capacitor discharges. The diode is included to ensure that current from the capacitor only flows through the resistor and LED, and does not head off towards the positive battery terminal. The circuit would work without the diode but it is good practice to get into the habit of using them where necessary.

Recall that electron flow is actually from the negative plate through the led through the resistor to the positive plate, although circuit diagram convention still uses conventional current.

The resistor does not have to be "before" the LED. Because they are connected in series, the same current passes through both components. Moving the resistor below the LED would provide the same current limiting. The LED's orientation matters, but the resistor's orientation and position in the series loop do not.

Electrons are already present throughout the entire circuit, including inside the resistor and LED. When the switch closes, the changing electric field propagates around the circuit and causes electrons everywhere to begin moving.

The resistor locally restricts charge movement. This produces a voltage drop across it and limits the current throughout the entire series circuit. Because charge cannot continuously accumulate between the LED and resistor, the same current must pass through both:

$$I_{\text{LED}} = I_{\text{resistor}}.$$

It is not necessary for electrons to enter the battery's positive terminal during the discharge phase, because the battery is no longer supplying the energy the capacitor is.

- **Note about the diode**: A diode's stated direction refers to conventional current, which is opposite to electron flow.
- **Reminder about conventional current**: Conventional current is a model that treats electricity as a flow of positive charge from the positive terminal to the negative terminal of a power source. Even though actual electrons flow in the opposite direction, this standard is used worldwide for circuit analysis and diagrams
- **Note about discharging**: Once the capacitor is discharged:
  - Excess electrons have left the negative plate.
  - Those electrons have entered the positive plate, filling its electron shortage.
  - Both plates are electrically neutral relative to each other.
  - The voltage across the capacitor is approximately 0 V.
  - Its electric field and stored energy are approximately zero.
  - It can no longer produce sustained current.
- **Measuring Capacitance**: The unit of capacitance is the farad (F). However, one farad is a lot of capacitance, so it is normal for capacitors to be measured in fractions of a farad such as microfarads (μF), which are millionths of a farad, nanofarads (nF), which are thousand-millionths of a farad, and picofarads (pF), which are million millionths of a farad.
- **The farad**: One farad is equal to one coulomb per volt. That is,

  $$F = \frac{C}{V} = \frac{Q}{V} = \frac{It}{V}.$$

  The C used here is for coulomb, although we usually use C for capacitance, so we can say

  $$C = \frac{Q}{V} = \frac{It}{v},$$

  where C is capacitance, Q is charge, V is voltage, and I is current.

- **Polarity**: Polarity is the property of having two opposite sides or ends (poles) with contrasting qualities, such as positive and negative electrical charges, while polarized describes something that has been restricted, separated, or aligned to possess or exhibit that distinct polarity or direction

  An electrical component is therefore **polarized** if it has **polarity**.

- **Types of Capacitor**: Capacitors come in different forms, sizes and styles, but they all contain at least two conducting plates that are separated by a dielectric. Capacitors can have a fixed value, for example, 1000 μF, or they can have a variable value. In variable capacitors, the amount by which the plates overlap can be varied, which allows the amount of charge stored to be varied. Variable capacitors can be used to tune a radio and are often called tuning capacitors.

  Fixed capacitors can be separated into two groups:

  - Polarized
  - Non-polarized

  Within each of these two groups there are sub-categories of capacitors:

  - Polarized
    - Electrolytic
    - Super-capacitors
  - Non-polarized
    - Ceramic
    - Vacuum, air, glass, silicon
    - Film

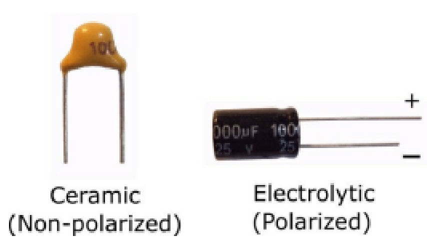


Figure 23 - Electrolytic and ceramic capacitors

- **Electrolytic and ceramic capacitors**: Polarized capacitors have one positive and one negative terminal and in order to work must be inserted into a circuit such that the positive terminal is connected to the positive side of the circuit, and the negative terminal is connected to the negative side. If the capacitor has one terminal that is longer than the other, the longer one is the positive terminal. Sometimes there is a '+' and '-' next to the terminals so that you can determine which terminal is which.

  Non-polarized capacitors can be inserted either way round.

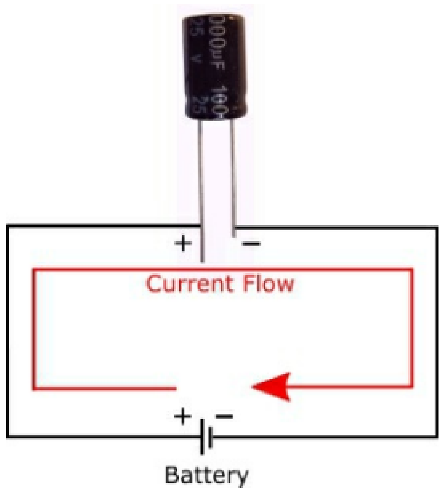


Figure 24 - Correct orientation of an electrolytic capacitor in a circuit

- **Capacitance Value, Voltage Rating and Tolerance**: Capacitors often have various letters and numbers written on them that let you identify the capacitance value, voltage rating and tolerance of the capacitor.

  Not all capacitors contain sufficient information printed on them to enable you to identify all the properties of the capacitor. Some capacitors are so small that there simply isn't enough room to contain all of the information. Hopefully though, when you bought your capacitor(s), the manufacturer will have supplied you with a data sheet to provide this information.

- **Capacitance Values**: Identifying the value of a larger capacitor such as an electrolytic capacitor is normally fairly easy because you can read the value written on the side of the component.

  Reading the capacitance value for a small ceramic capacitor is not quite so straight forward because you have to calculate the value based on the numbers printed on the capacitor.

  There are 3-digit numbers written on capacitors that enable you to work out their value in picofarads (pF).

1st Digit	2nd Digit	3rd Digit (Multiplier ×)	
0	0	0	×1
1	1	1	×10
2	2	2	×100
3	3	3	×1000
4	4	4	×10000
5	5	5	×100000
6	6	6	×1000000
7	7	7	×10000000
8	8	8	×100000000
9	9	9	×1000000000

Table 8 - Capacitor codes for calculating the value in picofarads (pF)

For example:

$$Digit1 = 1$$
$$Digit2 = 0$$
$$Digit3 = 4.$$

So, 104 is written on the capacitor. Therefore, the value is 10 x 10,000, which equals 100,000 pF, or 100 nanofarads (nF), or 0.1 microfarads (μF).

- **Voltage Rating**: This is the maximum voltage the capacitor is designed to handle. For larger capacitors it is written directly on the capacitor, but with smaller capacitors where space is limited, a code is used. Where space is so limited that there isn't even enough room for the code, you'll have to rely on the manufacturer's data sheet for the voltage rating.

Code	Maximum Voltage
1H	50 V
2A	100 V
2T	150 V
2D	200 V
2E	250 V
2G	400 V
2J	630 V

Table 9 - Capacitor voltage rating codes

- **Tolerance**: As with resistors, capacitors have a tolerance rating which means that the value written on the capacitor (or calculated if there isn't room to display the value) is probably not the accurate value.

Code	Tolerance
A	± 0.05 pF
B	± 0.1 pF
C	± 0.25 pF
D	± 0.5 pF
E	± 0.5%
F	± 1%
G	± 2%
H	± 3%
J	± 5%
K	± 10%
L	± 15%
M	± 20%
N	± 30%
P	-0 to +100%
S	-20 to +50%
W	-0 to +200%
X	-20 to +40%
Z	-20 to +80%

Table 10 - EIA codes for capacitor tolerances

- **Schematic Diagram Symbols**:

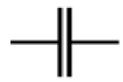
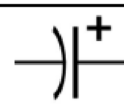
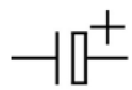
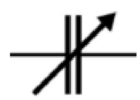
Component	Symbol	Identifier
Non-Polarized Capacitor		C, C1, C2, C3, and so on
Polarized Capacitor		
Polarized Capacitor		
Variable Capacitor		

Table 11 – Capacitor schematic symbols

Diodes

- **Intro**: A diode is a semiconductor device that passes current in one direction only.

  It is a polarized component that has two pins: the anode and the cathode. You can normally identify which is which because there will be some kind of marker on the diode such as a silver or colored band, which is next to the cathode.

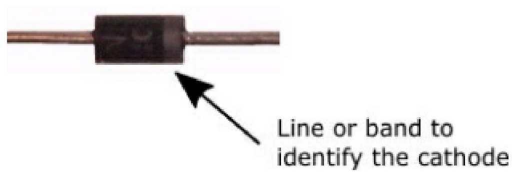
  

  Figure 27 - Identifying the cathode

  If the anode is connected to a higher voltage than the cathode (for example, the positive terminal of a battery), current flows through the component (from the anode to the cathode). If the cathode is connected to a higher voltage than the anode, current does not flow through the component.

  Different types of diode are available, which provides the ability for circuits to behave differently depending on current, voltage or frequency.

- **Electrode**: An electrode is an electrical conductor used to make contact with a non-metallic part of a circuit, such as an electrolyte, semiconductor, gas, or living tissue.
- **Anode**: An anode is the electrode where **oxidation** (the loss of electrons) occurs in an electrical or electrochemical device
- **Cathode**: A cathode is the electrode where a **reduction** reaction happens, meaning it is where chemical species gain electrons
- **Electrolyte**: An electrolyte in electronics is a substance that conducts electricity by moving charged particles called ions, rather than moving free electrons like metal wires do
  - **Ion Movement:** It contains positive and negative ions that flow to carry electrical charge.
  - **Blocking Electrons:** It stops electrons from passing directly through it.
  - **Forcing the Circuit:** By blocking electrons, it forces them to travel through an external circuit, which creates usable electric power.
- **Semiconductors**: A semiconductor is a substance that has a conductivity that lies somewhere between that of most metals and an insulator. Components made of a semiconductor substance - such as silicon - are essential in most electronic circuits.
- **How Diodes Work**: In its pure form, silicon is an insulator, but you can turn it into a semiconductor by doping. With doping, you mix a small amount of an impurity into the silicon, which changes its behavior. These impurities are classed as being either N-Type or P-Type.

- **N-type:** With N-Type doping, a small quantity of phosphorus or arsenic is added to the pure silicon. This causes some of the impurity's electrons to break free from their atoms, which are then able to move around within the silicon. As electrons have a negative charge, this causes this part of the diode to also have a negative charge. The N in N-Type stands for negative, and this side of the diode is the cathode.
- **P-Type:** With P-Type doping, a small quantity of boron or gallium is added to the pure silicon. The result of doing this is to cause holes to be created in the silicon's structure, where there is an absence of electrons, which creates a positive charge within the silicon. The P in P-Type stands for positive, and this side of the diode is the anode.

- **Dopant ions**: Dopant ions are charged impurity atoms intentionally added to a material to alter its electrical, optical, or structural properties

  They are atoms that has gained or lost an electrical charge and is embedded into a crystal lattice (such as silicon).

- **Holes**: If a silicon atom loses an electron, a hole is left behind where the electron used to be.

- **PN junction**: In a diode there are two doped layers: one N-Type and one P-Type. Where these two layers meet, a PN junction is formed. At this junction, the two materials cancel each other out and so a very thin layer is formed that is neither positively nor negatively charged. This layer is called the depletion layer, and no current can flow across it. However, when a voltage is applied across the PN junction, so that the P-Type anode is made positive and the NType cathode is made negative, current is able to flow. This is known as **forward bias**. If, however, the cathode is made positive and the anode negative - in other words the diode is inserted into the circuit the opposite way round - no current is able to flow. This is known as **reverse bias.**

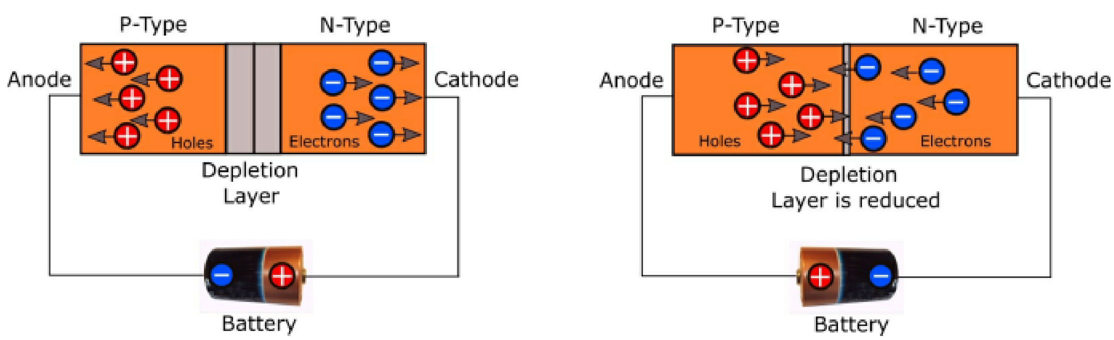


Figure 28 - How a diode works

When current is flowing through a diode, the voltage on the positive pin (anode) is higher than on the negative pin (cathode). This is called the diode's forward voltage drop and needs to be taken into account when trying to determine what size resistor you need to use

The diode depletion layer (or depletion region) is an insulating zone at the junction of p-type and n-type semiconductor materials where mobile charge carriers have recombined, leaving behind fixed ions with no free carriers to conduct current

A PN Junction Diode joins p-type material (filled with positive holes) and n-type material (filled with free electrons).

Electrons from the n-side cross into the p-side and combine with holes. This movement leaves behind positive ions on the n-side and negative ions on the p-side. These charges create an internal electric field that stops further diffusion, forming the depleted layer.

- **Forward Bias:** Connecting a positive voltage to the p-side and negative to the n-side shrinks the depletion layer, letting current flow past a threshold (about 0.7V for silicon).
- **Reverse Bias:** Reversing the voltage expands the depletion layer, blocking current flow completely

- **Mobile carriers**: Mobile charge carriers are particles, or particle-like vacancies, that can move through a material and transport electric charge. Their movement produces electric current.

  In a semiconductor, there are two kinds:

  - **Electrons:** Carry negative charge and move between available energy states in the material.
  - **Holes:** Carry effective positive charge. A hole is an empty position left by a missing electron.

  A hole is not a physical positive particle. It appears to move when nearby electrons repeatedly fill empty positions:

  For example, suppose there are several chairs in a row and one chair is empty. As people move into the neighboring empty chair, the empty chair appears to move in the opposite direction. A hole behaves similarly.

  - The N-type side has many mobile electrons.
  - The P-type side has many mobile holes.
  - The depletion region has very few of either, so it cannot conduct significant current.

  The depletion region also contains charged dopant ions, but those ions are locked into the silicon crystal and cannot move. Therefore, they are fixed charges, not mobile carriers.

- **Note about the depletion region**: the depletion region is not an empty physical void. The silicon atoms are still there; it is primarily a region with almost no mobile charge carriers.

- **More on reverse bias**:

  - Mobile holes in the P-type region are pulled toward the left terminal.
  - Mobile electrons in the N-type region are pulled toward the right terminal.
  - This leaves a wider region around the junction without mobile holes or electrons.

  The depletion region blocks current for two related reasons

  1. It has almost no mobile carriers to transport charge.
  2. Its electric field opposes carriers attempting to cross the junction.

- **Valence electrons**: The electrons found in the outermost shell (energy level) of an atom are called **valence electrons**.
- **More on dopants**: P-type silicon is commonly doped with an acceptor atom, such as boron. Boron has one fewer valence electron than silicon. It accepts an electron from the surrounding silicon structure:

  N-type silicon is commonly doped with a donor atom, such as phosphorus. Phosphorus has one extra valence electron, which it donates to the material

  The ionized dopant atoms left behind create the fixed charges in the depletion region.

  - On the P side, acceptor dopants accepted electrons and became fixed negative ions.
  - On the N side, donor dopants donated electrons and became fixed positive ions.

  Normally, mobile holes and electrons balance those dopant charges. Near the junction, however, the mobile electrons fill the mobile holes. This removes the balancing mobile carriers and leaves the dopant ions exposed:

  balance means "make the region electrically neutral."

  Electrons are abundant on the N side, so some diffuse into the P side. There, an electron can fill a hole.

  "Filling a hole" means that a free electron occupies the missing-electron position represented by the hole. The electron becomes bound in the silicon structure, so neither remains a mobile carrier

  The electron did not cease to exist; it simply stopped being mobile. The hole, the missing-electron position, no longer exists.

  Suppose three N-side electrons cross the junction and fill three P-side holes:

  - The N side has lost three mobile negative charges, leaving three positive donor ions uncompensated.
  - The P side has lost three mobile positive charges, leaving three negative acceptor ions uncompensated.

  "Exposed dopant ions" therefore means dopant ions whose charges are no longer balanced by nearby mobile carriers. They are not physically uncovered; their electrical charge is simply now uncompensated. These uncompensated ions produce the depletion region's electric field.

  During reverse bias, more mobile carriers are pulled away from the junction, exposing more dopant ions and making the depletion region wider.

- **Charge density**: Charge density is a measure of how much electric charge is accumulated over a specific length, surface area, or volume. It describes how tightly packed electric charges are within a given space. The SI unit for electric charge is the Coulomb (C),

- **Linear (λ):** Charge per unit length

$$\lambda = \frac{Q}{L}.$$

- **Surface (σ)**: Charge per unit area

$$\sigma = \frac{Q}{A}.$$

- **Volume (ρ):** Charge per unit volume

$$\rho = \frac{Q}{V}.$$

- **Charge neutrality (electroneutrality)**: Charge neutrality means that a macroscopic object or material has an equal amount of positive and negative charge, resulting in a net electric charge of zero

  The condition for charge neutrality in a system is

$$\rho_{\text{net}} + \rho_{+} + \rho_{-} = 0.$$

  It states that the net charge density (ρ_{net}) is equal to zero because the positive charge density (ρ_{+}) and the negative charge density (ρ_{-}) perfectly balance each other out.

- **Electric field with neutral charge**: If positive and negative charges are mixed together in equal amounts, their electric fields largely cancel:

  However, being neutral overall does not guarantee that the electric field is zero everywhere. For example, the depletion region contains equal total amounts of positive and negative fixed charge, but they are separated:

- **Types of Diode**: There are several types of diode available, with each type designed to perform a particular function in an electronic circuit.
  - **Regular Diode**: This type of diode is designed to stop electricity from flowing in the wrong direction. Whenever a diode is used in this book, it is a regular diode.
  - **Zener Diode**: This is a special kind of diode that allows current to flow when the diode is forward biased, but also allows current to flow when the diode is reverse biased so long as the voltage is above a certain value - the breakdown voltage - which is also known as the Zener voltage.
  - **Photo Diode**: This type of diode only conducts an electric current when it is exposed to light. A typical application might be a circuit in which a switch only operates when the diode is exposed to light.
  - **Schottky Diode**: This type of diode has a low forward voltage drop and can turn on and off very quickly when the breakdown voltage is reached, enabling it to respond very quickly in digital circuits.
  - **Light-Emitting Diode (LED)**: The best known type of diode is the light-emitting diode, or LED, which is a diode designed to give off light when a current passes through it.

- **Schematic Diagram Symbols**:

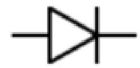
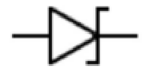
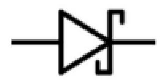
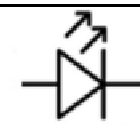
Component	Symbol	Identifier
Regular Diode		D, D1, D2, D3, and so on
Zener Diode		
Photo Diode		
Schottky Diode		
Light Emitting Diode (LED)		LED, LED1, LED2, and so on

Table 12 – Diode schematic symbols

Light Emitting Diodes (LEDs)

- **Intro**: A light-emitting diode - or LED - is a special type of diode that converts electrical energy into light

  LEDs are available in a range of colors and sizes and, as with other diodes, only allow current to flow in one direction. This means that when they are used in an electronic circuit, they need to be inserted in a particular way or they won't work.

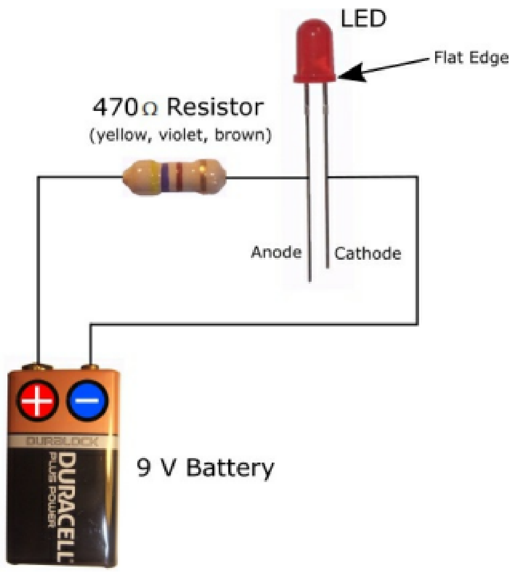


Figure 29 - An LED inserted into an electronic circuit

  If you look at the circuit in Figure 29 you can see that the LED has two pins, one of which is slightly longer than the other. The longer pin is the anode and has to be connected to the positive side of the circuit. The shorter pin is the cathode and has to be connected to the negative side of the circuit. When an LED is connected in this way it is forward biased. Current always flows from the anode to the cathode. There is also a flat edge on the cathode side of the LED, which enables you to identify which pin is which when the LED is inserted into a circuit.

  If you insert an LED the wrong way round in a circuit (reverse biased) you won't damage it - it just won't work.

- **Current and Resistance**: The more current that flows through the LED, the brighter it shines. Too much current will cause the LED to shine very brightly, but only for a short amount of time before it burns up. Most LEDs require about 20 mA of current to shine brightly, but without presenting any risk of damage to the LED.

  You can control how much current flows through an LED by using a resistor. If you use a 9 V battery as the power supply, you will need to use a resistor that is about 400 Ω to sufficiently protect the LED. If you use a battery with a lower voltage, you can get away with using a smaller resistor. If you use a resistor with a higher resistance value than is strictly necessary, the LED will be less bright than it would otherwise be, but it will still work.

- **Heat dissipation**: What a resistor does when it is connected in a circuit is to dissipate extra electrical energy (power) as heat energy. This means that if you use a resistor that is perhaps a bit on the

  big size (in terms of its resistance value) it may well start to get warm because it is having to dissipate a lot of electrical energy as heat in order to bring the current down to a level that is suitable for the LED. If this happens, try using a slightly smaller resistor. Don't go too small though, otherwise you'll run the risk that the resistor doesn't provide enough protection for the LED.

- **Forward voltage drop** V_F: When current is flowing through an LED, the voltage on the positive leg (anode) is higher than on the negative leg (cathode). This is called the LED's forward voltage drop (V_F) and is the voltage that is "lost" in the LED when it is operated at a certain current.

  Voltage drop is a reduction in voltage which occurs as electric current moves through a circuit, and is caused by the internal resistances of the various components in the circuit (resistors, connecting wire, LEDs, and so on).

  An LED does experience a fixed voltage drop when it turns on, and that drop in electrical energy is what creates light.

- **Electrical energy**: Electrical energy is a combination of voltage and current, so we can say that a voltage drop also causes a drop in the electrical energy in a circuit. This is important because it is the electrical energy that enables the circuit to do work, for example, illuminate an LED, drive a motor, and so on. If the energy lost in the circuit is too much before an LED is illuminated or a motor is driven, the circuit will fail to do its job properly.

  Electric power is the rate, per unit time, at which electrical energy is transferred by an electric circuit. The SI unit of power is the watt, which is equivalent to one joule per second. Electric power can be produced by electric generators or electric batteries

- **More on voltage drop**: Voltage drop is not necessarily a bad thing, and dropping voltage (and therefore electrical energy) across a motor is useful because in doing so, work is being performed by the motor. However, if voltage is dropped across a resistor, for example, this is undesirable because it means that electrical energy is being lost without doing anything useful. For a resistor, the electrical energy is lost in the form of heat energy

  Voltage drop in direct-current (DC) circuits is slightly different to that in alternating-current (AC) circuits.

- **Voltage drop in direct-current circuits**: Consider a circuit that is powered by a 9 V battery, which has one resistor and two LEDs all connected in series

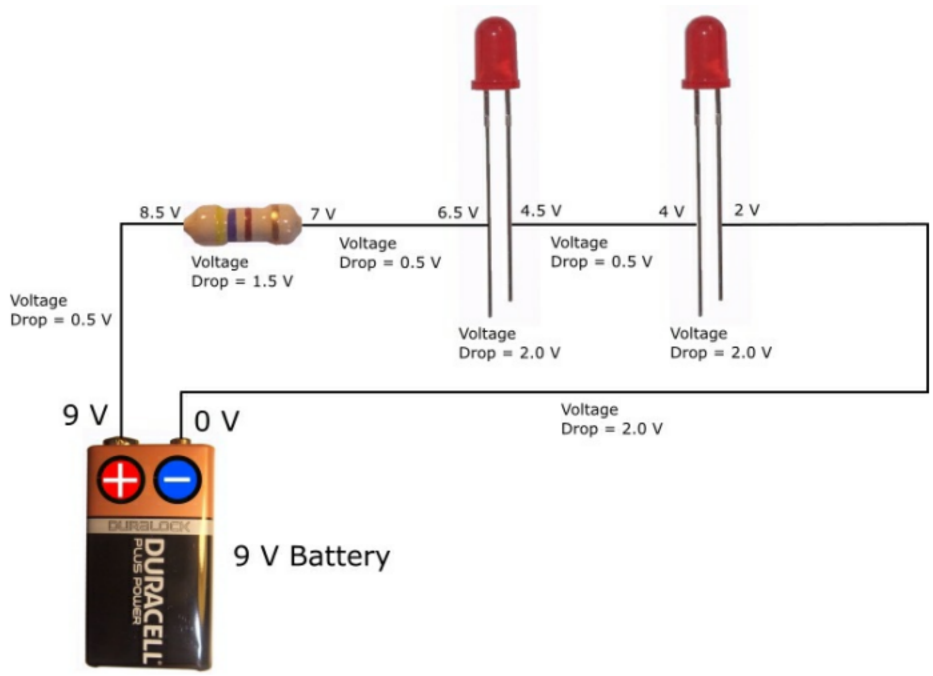


Figure 30 - "Sample " voltage drops in different parts of a circuit

The battery, the conductors (wires), the resistor, and the LEDs (the load) all have resistance and all use and dissipate supplied electrical energy to some degree. Their physical characteristics determine how much energy they use or dissipate. For example, the resistance of a wire depends upon the wire's length, its cross-sectional area, the type of material from which it is made, and its operating temperature.

If the voltage between the battery and the resistor is measured, the voltage potential (how much voltage is available) at the first resistor will be slightly less than 9 V. This is because as the current passes through the wire connecting the battery and the resistor, some of the electrical energy is lost due to the resistance of the wire, which means the energy is unavailable for the load (the LEDs). As the current passes through the resistor, more electrical energy is lost due to the resistance of the resistor. More electrical energy is lost as the current heads toward the first LED. As the current passes through the LED, more electrical energy is lost, but this time mainly in the form of light energy. This is good though because this is the purpose of the circuit - to illuminate this, and the second LED. The process of electrical energy being lost continues throughout the rest of the circuit until we finally get to the negative terminal of the battery at which point the voltage is 0 V.

The larger the resistor, the more energy is used by that resistor, and the bigger the voltage drop across that resistor.

Ohm's Law can be used to verify voltage drop. In a DC circuit, voltage is equal to the current multiplied by the resistance ($V = IR$). So, a 1 $k\Omega$ resistor will produce a larger voltage drop than a 47 Ω resistor, for a given current. Also, Kirchhoff's circuit laws state that in any DC circuit, the sum of the voltage drops across each part of the circuit is equal to the supply voltage. So, for the circuit shown in Figure 39, the sum of all the voltage drops is equal to the supply voltage of 9 V.

- **Types of LEDs**:

- Flashing LEDs, as the name suggests, flash when they are connected into a circuit. These LEDs contain a tiny integrated circuit (IC) that allows the LED to flash without requiring the use of an external IC such as a 555 Timer.
- RGB LEDs are even more impressive because they are able to glow in a range of colors - not just red, green and blue. RGB LEDs contain three LEDs (red, green and blue) in one package and, by mixing light from the three LEDs, you are able to produce other colors.

  RGB LEDs normally have four pins - one for each color plus a common. Depending on the LED, the common pin might be the anode or it might be the cathode. If you are including an RGB LED in a circuit, you need to know whether the common pin is the anode or the cathode, as well as the colors represented by the other pins. The RGB LED shown in Figure 32 is one I have in my electronics kit. Here, the common is the cathode, which means the red, green and blue pins are all anodes.

  RGB LEDs are different to other LEDs in that you cannot simply connect them to a battery (via a resistor of course) in order to get them to work. Instead, they require a pulse width modulation (PWM) digital signal as input on the red, green, and blue pins, which can be supplied by a micro controller or a 555 Timer. Depending on the LED, the common pin is then either connected to the positive (for a common anode) or negative (for a common cathode) supply.

Figure 32 - An RGB LED

- **Die**: A die in electronics is a small block of semiconducting material, usually silicon.
- **How an LED is Constructed**: The construction of an LED is quite different to that of a 'normal' diode. With an LED, the top of the device is surrounded by a protective, transparent, epoxy (plastic) case that protects the LED and helps to distribute the light generated by the LED.

  A metal cup (reflective cavity) is placed on the negative pin (Anvil) which holds a semiconductor die, which is a combination of two semiconductor materials – N type and P type - as well as an active region (known as the P-N junction) between them.

  The conical shape of the cup reflects the light emitted from the semiconductor die upwards. This is why LEDs appear brighter if you view them from above rather than the side. A wire bond connects the Post and the Anvil
- **How an LED Works**: The way an LED works is that when it is inserted into a circuit such that it is forward biased, electrons (-) and holes (+) start hopping backwards and forwards across the P-N junction. Whenever an electron finds and occupies a hole in an atom it makes the atom complete, which causes a tiny amount of energy to be released in the form of a photon of light.

  The photons are directed upwards towards the domed top of the LED's case which acts like a lens to concentrate the light.

Transistors

- **Intro**: Of all the electronic components we have looked at so far in this book, the transistor is the one that has had the biggest impact on electronics and the computer industry in general. The transistor is a semiconductor device, and is the building block of modern electronic systems. It can be used as a switch or to amplify electronic signals.

  Without transistors, many of the mobile devices we use today would not be mobile at all: laptops, mobile phones, and tablets would be too big to fit in a room in your house, let alone be portable.

- **Basic Operation of a Transistor**: A transistor is able to use a small signal that is applied between one pair of its pins (for example, the Base and Emitter) to control a larger signal across another pair of pins (for example, the Collector and Emitter). This property is referred to as gain, and it occurs when a stronger output signal (voltage or current) is produced that is proportional to a weaker input signal. In other words, the signal is amplified. Alternatively, a transistor can be used to turn current on or off in a circuit as an electrically controlled switch, where the amount of current is determined by other circuit components, for example, resistors.

  There are two types of transistors:

  - Bipolar Junction Transistor (BJT)
  - Field-Effect Transistor (FET)

  A bipolar junction transistor has pins labeled Base, Collector, and Emitter. A small current at the Base pin - flowing between the Base and the Emitter - can control or switch a much larger current flowing between the Collector and Emitter pins.

  For a field-effect transistor, the pins are labeled Gate, Source, and Drain, and a voltage at the Gate can control a current between Source and Drain. In this book we only use bipolar junction transistors, so any reference to transistors infers bipolar junction transistors.

- **How a Transistor is Constructed**: A transistor is similar to a diode except it has an additional semiconductor connected to one end, which can be either N-Type or P-Type. Therefore, instead of just having one P-N junction, a transistor has two. This gives rise to two possible semiconductor combinations:

  1. PNP transistors
  2. NPN transistors

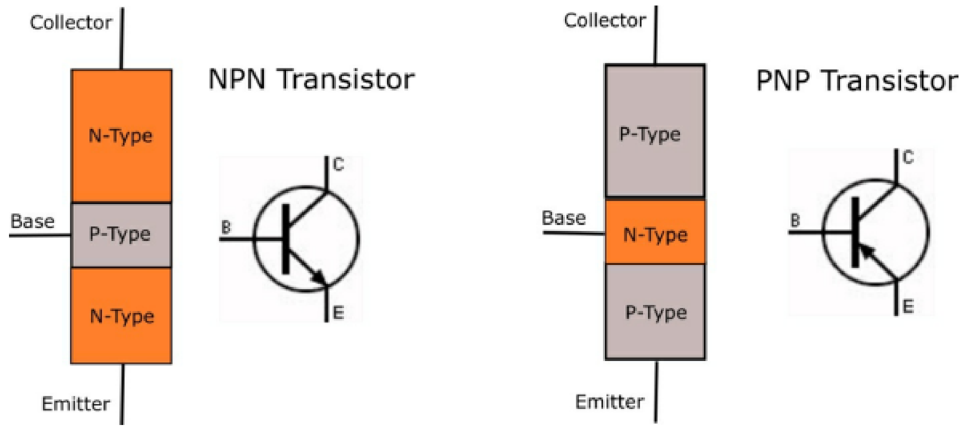


Figure 36 - NPN and PNP transistors

- **NPN and PNP Transistors**: A transistor, whether it is an NPN or a PNP transistor, will typically look like the image shown in Figure 37.

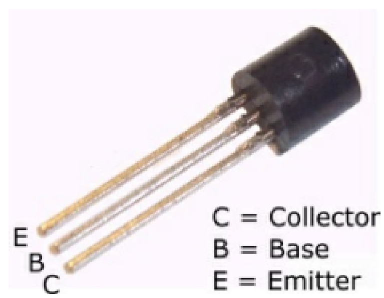


Figure 37 - A typical transistor

The way that the pins are labeled in Figure 37 is not true for all transistors. The best way to identify which pin is which is to refer to the data sheet supplied with the transistor(s) if you can.

Although NPN and PNP transistors are constructed differently and have to be connected up differently in a circuit in order for them to work, they both do the same thing: they allow for amplification and/or switching. If you design a circuit that uses, say, an NPN transistor, it is quite often easy to change it to use a PNP transistor

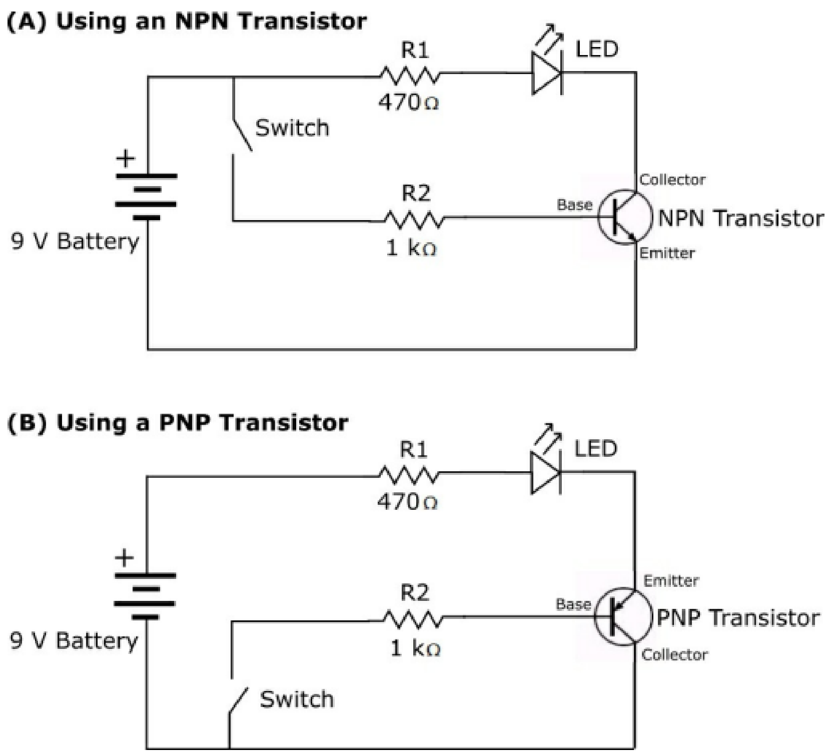


Figure 38 - Switching circuit that uses (A) an NPN transistor, and (B) a PNP transistor

Although the two circuits do the same thing and look very similar, it is important to note the differences in the way that the transistors are connected up.

Due to them being internally constructed differently, the way that voltage and current is allocated within PNP and NPN transistors is different. An NPN transistor needs a positive voltage at both the Collector and the Base, whereas a PNP transistor needs a positive voltage at the Emitter and a more negative voltage (than at the Emitter) at the Base.

Since voltage allocation is different between the two types of transistor, the way current is used to switch them on is also different. An NPN transistor is switched on when sufficient current is supplied to the Base. Therefore, the Base of an NPN transistor must be connected to a positive voltage for current to flow into the Base. A PNP transistor is the opposite; current flows out of the Base by giving it a more negative (a lower) voltage than is supplied to the Emitter.

Another thing that is different between NPN and PNP transistors is that the direction of their output current is different. In an NPN transistor, output current flows from the Collector to the Emitter, whereas in a PNP transistor, output current flows from the Emitter to the Collector.

- **How NPN and PNP Transistors are Switched On and Off**:
  - **NPN Transistor**: As you increase current to the Base, the transistor is turned on more and more until it conducts fully from the Collector to Emitter.

As you decrease current to the Base, the transistor turns on less and less, until the current is so low that the transistor no longer conducts across the Collector to the Emitter, and switches off.

Current does not flow from the collector to the emitter when no current is supplied to the base

The base acts like an ON/OFF switch for the transistor. When base current is zero, the transistor is turned completely off.

  - **PNP Transistor**: As current flows from the Base toward the negative battery terminal, the transistor is on and conducts current across the Emitter and Collector.

- **Identifying the Pins**: Although transistors come in different shapes and sizes, they all have three pins. Identifying which is which is not always easy though, so the best course of action when you need to identify them is to refer to the data sheet supplied with the transistor if you can. If you don't have the data sheet available, the following table might be helpful.

For **NPN** transistors:

Transistor Name	Pins 1 2 3
BC 546, 547, 548, 549, 550, BC 337, AC 187	CBE
TIP 120, 121, 122	BCE
BD 139	ECB
BF 494, 495	CEB
C2570	BEC
C1730	ECB
BD 233	ECB
BD 647, 677	BCE
D882 / 2SD882	ECB
D313 / MJE 13005	BCE
D44VH10	BCE
SF 245	CBE
BUF 742	BCE

For **PNP** transistors:

Transistor Name	Pins 1 2 3
2N 2222A, 2N 3904	EBC
TIP125, 126, 127	EBC
BD140	ECB
MPSA 92, 42, 44	EBC
BC 636	ECB
SK / CK / BEL 100P	EBC
AC188	EBC
BC 556 B	CBE
BC 557, 558	EBC

- **Schematic Diagram Symbols**

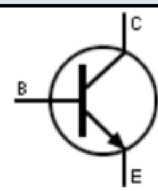
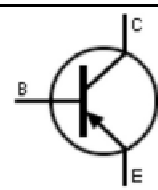
Component	Symbol	Identifier
NPN Bipolar Transistor		Q, Q1, Q2, and so on
PNP Bipolar Transistor		

Table 14 - Transistor schematic symbols

Integrated circuits

- **Intro**: So far in this book we have looked at individual electronic components such as transistors, diodes and resistors. These are known as discrete components and by using them together we are able to build electronic circuits that do different things. It is possible, however, to miniaturize these discrete components and combine lots of them into a single electronic component called an integrated circuit (IC) that can perform a particular task or tasks. ICs - also called silicon chips or just chips - can be complicated, as is the case with a random access memory (RAM) chip, or they can be relatively simple, as is the case with a 555 Timer (the subject of the next chapter). ICs can themselves be combined with other ICs and discrete components to create complete electronic systems, for example, a computer's system (mother) board.

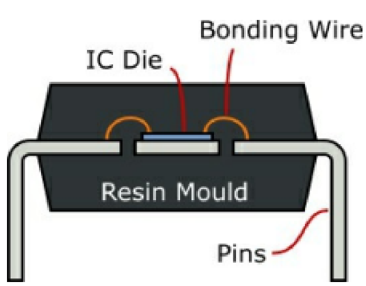
  An IC is built from a single piece of silicon crystal with all the individual components embedded directly into the crystal. The number of transistors contained in a modern-day IC can be truly staggering, with some microprocessors containing billions of them. At the other end of the scale, the 555 Timer contains about 25 transistors, along with a few other components.

- **How Integrated Circuits are Packaged**: Although there a few ways in which ICs can be packaged, the most common package - and the only one we examine in this book - is the dual inline package (DIP).

Figure 39 - A 555 Timer packaged in a DIP

  Looking at Figure 39 we can see that a DIP IC comprises a rectangular case and two rows of pins. The IC die itself is housed within the case, which is normally made of some kind of plastic resin, but can be made of ceramic.

  The pins protrude from the longer sides of the package along the seam and are bent downward at 90 degrees



Inside the package, the pins are embedded in the lower half. At the center of the package is a rectangular space into which the IC die is cemented. Ultrafine bond wires are used to connect the die and the pins. If a single bond wire breaks or detaches, the entire IC may become useless. The top of the package covers this delicate assembly, protecting it from damage or contamination by foreign materials.

- **Pin Identification**: All the pins in a DIP are numbered. In order to be able to identify which pin is which, there is normally a mark of some sort next to pin 1

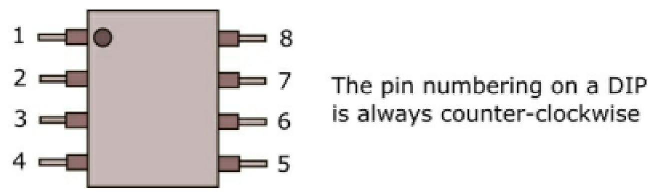


Figure 41 - Pin numbering on an 8-pin DIP

Although the pin numbering on a DIP is as shown in Figure 41, when an IC is shown in a schematic diagram the positioning of the pins may well be different. For example, when a 555 Timer is shown in a schematic diagram the pins are (almost) always shown in a particular way, and any unused pins are not shown at all. We'll see more of this when we look at the 555 Timer in the next chapter.

The 555 Timer

- **Intro**: Introduced in 1971, and with an estimated 1 billion (or more) being produced every year, the 555 Timer is quite possibly the most popular IC in the world.

  The circuit contained within the 555 Timer chip is called a **multivibrator**, which is used to implement a variety of simple two-state devices such as:

  – **Relaxation oscillators**: a circuit that produces a repetitive output signal, such as a square wave. This could be used to cause a warning light to flash on an off, for example. When used in this way, the 555 Timer is said to be in Astable mode.

    Astable means no stable state.

  – **Timers**: a circuit enabling a light to be switched on for a certain amount of time, for example. When used in this way, the 555 Timer is said to be in Monostable mode.

    Monostable means one stable state.

  – **Flip-flops**: a circuit that has two stable states, for example, on receipt of a trigger, the circuit can permanently produce an output of +5 V until a second trigger causes the output to go to 0 V. It will then stay at 0 V until a third trigger is received, which will cause the output to go back to +5 V, and so on. Flip-flops are one of the fundamental building blocks of digital electronics systems used in computers and communications systems. When used in this way, the 555 Timer is said to be in Bistable mode

    Bistable means two stable states.

  The output from a 555 Timer typically looks like that shown below in Figure 42. This is a digital (rectangular) signal, which can be either off (0 V), or on (say 3 V). How long the signal is on or off is determined by the timing mechanism inside the 555 Timer's circuit.

Figure 42 - Typical output from a 555 Timer

- **Analog and digital**: Analog and digital are two ways to send, store, and process information using continuous waves or discrete numbers

  – **Analog:** Uses continuous, smooth signals (like a sine wave) that can take any value in a range. Think of turning a volume knob or looking at a clock with hands.

  – **Digital:** Uses separate, step-like values (like a square wave) represented as binary numbers (1s and 0s, or on and off). Think of flipping a light switch or reading a digital watch

Analog circuits process continuously changing voltage or current signals, while digital circuits process information using discrete, on-off binary states (1s and 0s)

- **555 timer**: The 555 Timer is an analog-digital device. The output, as shown above, is a digital signal, but the input to the chip is an analog signal produced by a resistor-capacitor (RC) circuit. It is this circuit that determines how long the output signal is high or low, which can be varied by using different value resistors and capacitors.

  Some devices such as LEDs work quite happily when a continuous DC signal is applied to them. Other devices, however, such as a servo motor or a piezo buzzer require a digital signal as input.

- **555 Timer Pins**: The pins on the 555 Timer are as shown below in Figure 43.

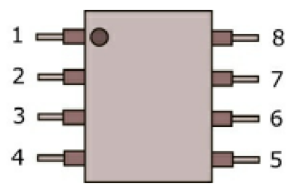
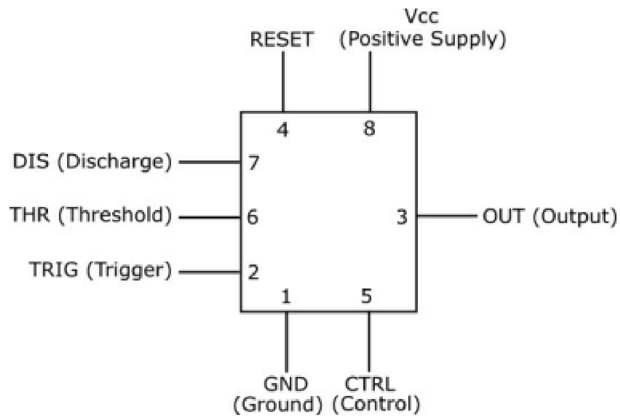


Figure 43 - Pin numbering on the 555 Timer

555 Pin#	Pin name	Pin purpose
1	GND	Ground supply: This is the ground reference voltage (zero volts). You connect this pin to the negative terminal of the battery.
2	TRIG	Trigger: The OUT pin (pin 3) goes high and a timing interval starts when the voltage on this (TRIG) pin is kept at a low voltage. The low voltage needs to be less than 1/2 of the CTRL voltage, which is typically 1/3 of the supply voltage applied to pin 8 (Vcc)
3	OUT	Output: This is where the digital output signal leaves the chip. The output is either very low (almost 0 V) or close to the supply voltage (Vcc) applied to pin 8. How long the output signal is high or low depends on the connections made to pins 2, 4, 5, 6, and 7.
4	RESET	Reset: A timing interval may be reset by driving this input to GND, but the timing does not begin again until RESET rises above approximately 0.7 volts (triggered by pin 2). In order for the 555 Timer to operate, this pin must be connected to the supply voltage (Vcc). This pin overrides TRIG, which in turn overrides THR (unless you are using an LM555 Timer, in which case THR overrides TRIG).
5	CTRL	Control: This pin provides "control" access to the 555 Timer's internal voltage divider. By applying a voltage to this pin you can alter the timing characteristics of the device. However, in most applications, this pin is not used and is connected to ground via a small (0.01 µF) decoupling capacitor that is used to smooth out any fluctuations in the supply voltage that may affect the operation of the timer.
6	THR	Threshold: The purpose of this pin is to monitor the voltage across the capacitor that is discharged by pin 7 (DIS). When the voltage reaches 2/3 of the supply voltage (Vcc), the output on pin 3 goes low, ending the timing cycle.
7	DIS	Discharge: This pin is used to discharge an external capacitor that is working in conjunction with a resistor to control the timing interval.
8	VCC	Positive supply: This pin is connected to the positive supply voltage, which must be in the range 4.5 to 15 V. The 9 V battery supply we use in the circuits in this book is ideal, but you could equally use four 1.5 V AA or AAA batteries to obtain a supply voltage of 6 V.

- **How the Pins on a 555 Timer are Normally Shown in a Schematic Diagram**: When a 555 Time is shown in a schematic diagram, the pins are not positioned as shown above in Figure 43. Instead they are almost always shown as in Figure 44 below.



- **A Look Inside the 555 Timer**: Depending on the manufacturer, the 555 Timer contains the equivalent of 25 transistors, about 16 resistors and two diodes. Figure 45 shows a functional block diagram of the NE555 Timer, which is one of the many 555 Timers on the market.

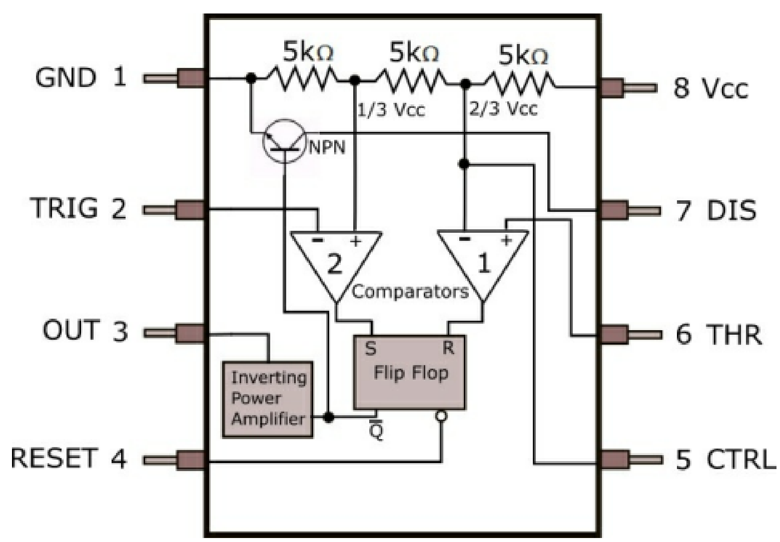
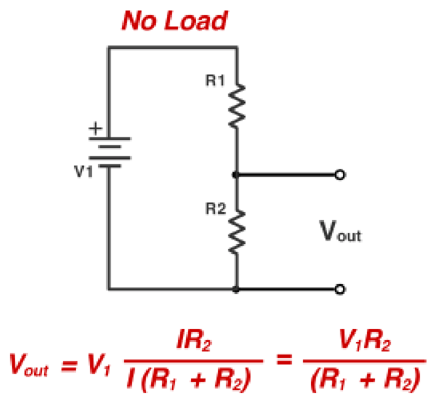


Figure 45 - Functional block diagram of the NE555 Timer

- **Voltage divider**: A voltage divider is a simple circuit that produces an output voltage smaller than its input voltage. It uses two resistors connected in series:

$$V_{\text{in}} \to R_1 \to V_{\text{out}} \to R_2 \to \text{GND}.$$



Because the resistors are connected in series, they share the same amount of current.

$$I = \frac{V}{R} = \frac{V_{\text{in}}}{R_1 + R_2}.$$

The output is the voltage across R_2

$$V_{\text{out}} = IR_2.$$

Thus,

$$V_{\text{out}} = \frac{V_{\text{in}} R_2}{R_1 + R_2}.$$

The output voltage is the electrical potential measured between the middle of the two resistors and ground.

Suppose $V_{in} = 12$ V. The top of R_1 is therefore at 12 V, while the bottom of R_2 is at 0 V because it is connected to ground. The voltage must decrease from 12 V to 0 V across the two resistors.

Because the same current flows through both resistors, each resistor receives part of that total voltage drop:

The output voltage is the voltage remaining at the midpoint. It is also equal to the voltage across R_2:

The output terminal is simply a point where another circuit can measure or use that modified voltage.

We add the resistor values because R_1 and R_2 are connected in series. The current must pass through R_1 and then R_2, so both resistors oppose the same current.

- **Connecting batteries positive directly to its negative**: If you connect a battery's positive terminal directly to its negative terminal, you create a short circuit. The voltage does not simply disappear harmlessly, the battery drives a very large current through the small resistances of the wire and battery.

A real circuit always has some resistance:

$$R_{\text{total}} = R_{\text{wire}} + R_{\text{internal}}.$$

Thus,

$$I = \frac{V_{\text{battery}}}{R_{\text{wire}} + R_{\text{internal}}}.$$

Because those resistances are very small, the current can be extremely large.

Voltage decreases across resistance according to:

$$V_{\text{drop}} = IR.$$

Even a wire has a small resistance. However, the battery's internal resistance is often much greater than the short wire's resistance, so most of the voltage drop occurs inside the battery, with smaller drops across the wire and connection points.

As the current moves through these resistances, electrical energy becomes heat:

$$P = I^2 R.$$

This can make the wire and battery dangerously hot.

- **Three voltage divider**: A three-resistor voltage divider consists of three resistors connected in series:

  $$V_{\text{in}} \to R_1 \to V_1 \to R_2 \to V_2 \to R_3 \to \text{GND}.$$

  The current is the same across all three components:

  $$I = \frac{V_{\text{in}}}{R_1 + R_2 + R_3}.$$

  Measured relative to ground, the two junction voltages are:

  $$V_1 = V_{\text{in}} \frac{R_2 + R_3}{R_1 + R_2 + R_3}$$
  $$V_2 = V_{\text{in}} \frac{R_3}{R_1 + R_2 + R_3}.$$

  In general,

  $$V_{\text{tap}} = V_{\text{in}} \frac{\text{total resistance below the tap}}{\text{total resistance in the divider}}.$$

- **Entering and exiting a junction**: A junction is simply a point where components are connected by conducting wire. For two series resistors, the middle junction is the wire connecting them:

  $$R_1 \to \text{junction} \to R_2.$$

  "Entering the junction" means charge is flowing into that connecting wire from R_1. "Exiting the junction" means charge is flowing out of it through R_2.

- **Electrical node**: An electrical node is a connection point or junction in a circuit where two or more circuit components or wires meet

- **Stable series circuit**: A stable series circuit is a closed electrical circuit where components are connected end-to-end in a single path, operating under conditions where the current, voltage, and component temperatures remain steady over time
- **Accumulating charge changes voltage**: Since

  $$Q = CV \implies V = \frac{Q}{C}.$$

  If charge increases, voltage increases. Voltage describes the electrical potential created by charge. When net charge accumulates on a junction, that charge produces an electric field, which changes the junction's electrical potential, its voltage.
- **Conservation of charge**: Current measures the rate at which charge flows:

  $$I = \frac{\Delta Q}{\Delta t}.$$

  Conservation of charge means electrical charge cannot simply appear or disappear. Therefore, any charge flowing into the junction must do one of two things:

  1. Flow out of the junction.
  2. Accumulate at the junction.

  $$I_{\text{in}} - I_{\text{out}} = \frac{dQ_{\text{junction}}}{dt}.$$

  Suppose 5 mA enters a junction while only 3 mA exits. The remaining 2 mA accumulates there:

  Because 1 mA = 1 millicoulomb per second, the junction would gain 2 millicoulombs of charge every second.

  However, charge accumulation changes the junction's voltage. That voltage change alters the electric field and therefore the currents through the resistors. The adjustment continues until the outgoing current equals the incoming current:

  $$I_{\text{in}} = I_{\text{out}}.$$
- **Why is current the same?**: The current is the same because a series circuit provides only one path for charge to travel.

  Consider the connection between two resistors:

  $$R_1 \to \text{junction} \to R_2.$$

  If 5 mA entered the junction but only 3 mA left, charge would accumulate there at 2 mA. That accumulated charge would change the junction's voltage and electric field until the outgoing current matched the incoming current.

  This is expressed by conservation of charge:

  $$I_{\text{in}} - I_{\text{out}} = \frac{dQ}{dt}.$$

  In a steady-state series circuit, charge is not continuously accumulating at any junction, so:

  $$\frac{dQ}{dt} = 0.$$

  Thus,

  $$I_{\text{in}} = I_{\text{out}}.$$

- **Adding the resistor values**: Why do we add $R_1 + R_2 + R_3$? Recall that the same current passes though each component, and the sum of the voltage drops must equal the total voltage.

  $$V_{\text{in}} = V_1 + V_2 + V_3.$$

  So,

  $$V_{\text{in}} = IR_1 + IR_2 + IR_3 = I(R_1 + R_2 + R_3).$$

  Thus,

  $$I = \frac{V_{\text{in}}}{\sum_i R_i}.$$

- **Differential inputs**: Differential signalling is a method for electrically transmitting information using two complementary signals. The technique sends the same electrical signal as a differential pair of signals, each in its own conductor.

  Electrically, the two conductors carry voltage signals which are equal in magnitude, but of opposite polarity. The receiving circuit responds to the difference between the two signals, which results in a signal with a magnitude twice as large.

- **Gain**: Gain describes how much a circuit increases or decreases a signal. Voltage gain is the ratio of output voltage to input voltage:

  $$A_v = \frac{V_{\text{out}}}{V_{\text{in}}}.$$

  For example, if an amplifier receives 0.2 V and produces 2 V:

  $$A_v = \frac{2}{0.2} = 10.$$

  The circuit has a voltage gain of 10.

  Gain has no unit because it is one voltage divided by another voltage.

- **op-amp (operational amplifier)**: An operational amplifier, usually called an op-amp, is an electronic component that amplifies the difference between two input voltages.

  It has two inputs

  - **Non-inverting input**: V_+
  - **inverting input**: V_-

  its basic equation is

  $$V_{\text{out}} = A(V_+ - V_-).$$

  Here, A is the op-amp's open-loop gain, which is typically extremely large.

  Open-loop gain is the op-amp's gain when there is no feedback connection from the output to either input. It is written as

  $$A_{OL}.$$

  V_+ and V_- are simply names for the voltages measured at the op-amp's two input pins. They only identify the two different input pins.

The + input is called non-inverting because increasing its voltage pushes the output upward. The − input is called inverting because increasing its voltage pushes the output downward.

- Upward means the output voltage becomes more positive.
- Downward means the output voltage becomes less positive or more negative.

If V_+ is slightly greater than V_-, the output moves toward the positive supply voltage.

If V_- is slightly greater than V_+, the output moves toward the negative supply voltage, or toward ground in a single-supply circuit.

For example, suppose the op-amp's gain is 100,000, and

$$V_+ - V_- = 0.001V.$$

Then, the theoretical output would be

$$V_{\text{out}} = 100000(0.001) = 100V.$$

However, an op-amp cannot produce 100 V when powered by only 5 V. Its output stops near the supply limit. This is called **saturation.**

A typical op-amp has at least five important connections:

- Positive power supply
- Negative power supply or ground
- Non-inverting input
- Inverting input
- Output

An operational amplifier (op-amp) is called "operational" because it was originally designed to perform mathematical operations (like addition, subtraction, integration, and differentiation) in analog computers, not because its output voltage exceeds its input voltage

- **Negative feedback**: Op-amps are normally used with negative feedback, where some of the output is connected back to the inverting input.

Negative feedback causes the op-amp to adjust its output until:

$$V_+ \approx V_-.$$

This is sometimes called the virtual-short rule. The inputs are not physically connected, but their voltages become almost equal while the op-amp is operating normally with negative feedback.

If V_{out} is connected directly by a wire to the V_- input, both points belong to the same electrical node. An ideal wire has zero resistance. Using Ohm's law, the voltage difference across it is:

$$\Delta V = IR.$$

Since an ideal wire has $R = 0$,

$$\Delta V = I(0) = 0.$$

Therefore, there can be no voltage difference between the two ends of the wire:

Two useful approximations for an ideal op-amp are:

$$V_+ \approx V_-$$
$$I_+ \approx I_- \approx 0.$$

The second rule means that practically no current enters the op-amp's input terminals.

- **Voltage follower**: The simplest negative-feedback circuit connects the output directly to the inverting input, while the signal enters the non-inverting input.

  Because feedback makes $V_- \approx V_+$:

  $$V_{\text{out}} \approx V_{\text{in}}.$$

  This circuit has a gain of one

  $$A_v = \frac{V_{\text{out}}}{V_{\text{in}}} = 1.$$

  Although it does not increase the voltage, it acts as a **buffer**. It prevents a load from drawing significant current from the original signal source.

- **Differential gain of an op-amp**: An op-amp does not directly amplify just one input voltage. It amplifies the difference between its two inputs:

  $$V_{\text{out}} = A(V_+ - V_-).$$

  The difference

  $$V_d = V_+ - V_-$$

  is called the **differential input voltage**. Thus,

  $$V_{\text{out}} = AV_d.$$

- **Open-loop gain**: Open-loop gain is the op-amp's gain when there is no feedback connection from the output to either input (A_{OL}). The open-loop equation is

  $$V_{\text{out}} = A_{OL}(V_+ - V_-).$$

  The open-loop gain is intentionally very large so that negative feedback can control the op-amp accurately.

  $$V_+ - V_- = \frac{V_{\text{out}}}{A_{OL}}.$$

  If A_{OL} is very large, the op-amp needs only a tiny difference between its inputs to produce the required output.

  An op-amp contains several internal transistor amplification stages. If one stage has a gain of 100 and another stage has a gain of 1,000, their combined gain is approximately:

  $$100 \times 1000 = 100,000.$$

- **Closed-loop gain**: Most amplifier circuits connect the output back to an input. This is called feedback, and the resulting gain is called the closed-loop gain. Negative feedback sacrifices the op-amp's enormous open-loop gain to obtain a smaller, stable, and predictable circuit gain.

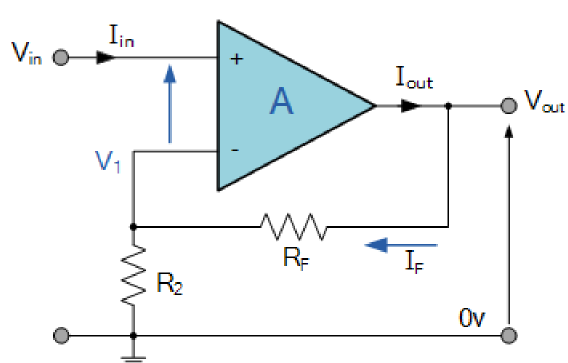
For example, resistors might configure an op-amp for a closed-loop gain of:

$$A_{CL} = 10.$$

- **What is a signal**: A signal in an electronic circuit is a time-varying voltage or current that carries information from one place to another
  - **Analog Signals**: These use a continuous, smooth range of values. The voltage or current changes fluidly over time. Sound waves from a guitar or temperature readings from a sensor are analog because they can be any value in between a minimum and maximum limi
  - **Digital signals**: These use distinct, separate levels to send data, usually restricted to just two states (on and off, or high and low). A high voltage represents a binary 1 (true), and zero volts or ground represents a binary 0 (false). Computers and microcontrollers use digital signals to process code and math.
- **Amplifying a signal**: Amplifying an input signal in electronics means increasing its voltage, current, or overall power while keeping the original shape and information of the signal as exact as possible.
- **Signal polarity**: Input signal polarity in electronics is the positive or negative orientation of a voltage or current waveform relative to a common reference or ground point
- **Signal phase**: Two signals are **in phase** when their peaks (highest points) and troughs (lowest points) occur at the exact same time.

  Two signals are **out of phase** (specifically 180 degrees out of phase) when the peak of one signal aligns perfectly with the trough of the other. One goes up exactly when the other goes down.
- **Feedback signal**: A feedback signal in electronic circuits is a portion of the output signal that is routed back to the input to control, regulate, or modify the circuit's behavior
- **Positive and negative feedback**: Positive feedback happens when the feedback signal is in phase with the original input signal. It adds to the input, which grows the output until the circuit reaches its maximum limit or saturates

  Negative feedback happens when the feedback signal is out of phase (180 degrees opposite) with the input. It subtracts from the input, which counteracts changes and stabilizes the system.
- **Non-inverting op-amp**: A non-inverting op-amp amplifies an input voltage without reversing its polarity.



In the diagram, V_{in} is connected to the op-amp's + input, while R_F and R_2 form a negative-feedback network.

The op-amp produces

$$V_{\text{out}} = A(V_+ - V_-).$$

where A is its very large open-loop gain. Because the output is fed back to the input, the circuit continually adjusts V_{out} until approximately

$$V_- \approx V_+ = V_{\text{in}}.$$

The feedback resistors form a voltage divider. Since

$$\begin{aligned} V_{\text{out}} &= V_{R_F} + V_{R_2} = I_F R_F + I_F R_2 \\ &= I_F(R_F + R_2), \end{aligned}$$

we have

$$I_F = \frac{V_{\text{out}}}{R_F + R_2}.$$

Notice that

$$V_- = I_F R_2.$$

Thus,

$$V_- = V_{\text{out}} \frac{R_2}{R_F + R_2}.$$

Since $V_- \approx V_+ = V_{\text{in}}$,

$$V_{\text{in}} \approx V_{\text{out}} \frac{R_2}{R_F + R_2}.$$

Hence,

$$V_{\text{out}} = V_{\text{in}} \left(1 + \frac{R_F}{R_2}\right).$$

The gain is then

$$A = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{R_F}{R_2} + 1.$$

The main point of a non-inverting op-amp is to amplify an input signal while keeping its original polarity and phase.

- **Signal attenuation**: Signal attenuation is the gradual loss of strength, amplitude, or power that a signal experiences as it travels through a medium like a cable, fiber optic strand, or the air
- **Summing points**: A summing point in electronics and control systems is a node or block diagram symbol where two or more input signals are combined algebraically by addition or subtraction.
- **Ideal op-amp**: For an ideal op-amp, the input resistance is infinite, so

  $$I_{\text{op-amp}} = 0.$$

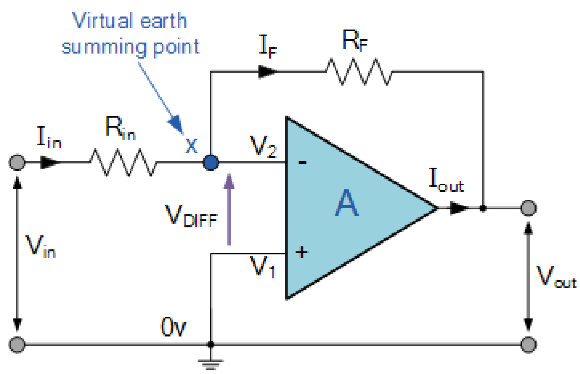
  Kirchoff's current law says that current cannot accumulate at the junction, so

  $$I_{\text{in}} = I_f + I_{\text{op-amp}} \implies I_{\text{in}} = I_{\text{f}}.$$

- **Inverting op-amp**: The inverting operational amplifier circuit is the simplest and most commonly used op-amp topology where its output voltage changes in the opposite direction to its input voltage

An Inverting Operational Amplifier (inverting op-amp) is basically fixed-gain voltage amplifier circuit configuration that amplifies an input signal while "inverting" (hence the name) its output signal by 180 degrees compared to its input.

That means a positive input signal produces a negative output voltage signal and vice-versa. In other words, the output is 180o out-of-phase with the input, producing "inversion".



Because of the feedback,

$$V_- \approx V_+.$$

Because V_+ is grounded,

$$V_+ = 0.$$

Consequently,

$$V_- \approx 0.$$

This point is called a **virtual ground**. It has approximately $0V$, but it is not physically connected to ground.

The input current through R_{in} is

$$I_{\text{in}} = \frac{\Delta V}{R} = \frac{V_{\text{in}-0}}{R_{\text{in}}} = \frac{V_{\text{in}}}{R_{\text{in}}}.$$

Ideally, no current flows into the op-amps input terminal. Hence,

$$I_f = I_{\text{in}}.$$

This is due to Kirchhoffs's circuit law, which states that for any junction, the sum of currents flowing into that node is equal to the sum of currents flowing out of that node.

$$\sum_{i=1}^{n} I_i = 0,$$

where n is the total number of branches with currents flowing towards or away from the node. Since I_{in} flows into the summing point, and I_{f}, $I_{\text{op-amp}}$ flow out,

$$I_{\text{in}} - I_f - I_{\text{op-amp}} = 0 \implies I_{\text{in}} = I_f.$$

We remark that if currents entering a junction are positive, currents leaving must be negative.

Now,

$$I_f = \frac{\Delta V}{R} = \frac{0 - V_{\text{out}}}{R_f} = -\frac{V_{\text{out}}}{R_f}.$$

Thus,

$$\frac{V_{\text{in}}}{R_{\text{in}}} = -\frac{V_{\text{out}}}{R_f}.$$

Hence,

$$V_{\text{out}} = -\frac{R_f}{R_{\text{in}}} V_{\text{in}},$$

and

$$A = -\frac{R_f}{R_{\text{in}}}.$$

- **Recall negative voltage**: A negative voltage simply means that a point is at a lower electrical potential than the chosen 0 V reference. Voltage is always measured between two points. So the voltage is below ground.

  Conventional current through a resistor flows from higher voltage toward lower voltage:

  $$+5V \to 0V \to -5V.$$

  - If $V_{out} = -5$ V, current through R_f flows from the 0 V summing point toward the output.
  - If $V_{out} = +5$ V, current flows from the output toward the summing point.

- **Sink current**: Sink current is the electric current that flows into an output or input pin of a device (like a microcontroller, logic gate, or integrated circuit) and travels toward the ground.

- **Sourcing current**: When a device sources current, it acts as the power supply, pushing electric current out of its output pin and into an external load.

- **What this implies for the inverting op-amp**: For a positive input voltage, the output of an inverting op-amp becomes negative. In that case, conventional current flows through the feedback resistor from the summing junction toward the output.

  Because V_{out} is negative, I_f is positive in the direction shown:

  $$\text{input source } \to R_{\text{in}} \to \text{ summing junction } \to R_f \to \text{ op-amp output.}$$

  The op-amp is therefore sinking conventional current through its output terminal. Internally, that current is carried toward the op-amp's negative power-supply rail.

  The feedback is still called negative feedback even though current flows toward the output. "Feedback" describes how the output voltage affects the input error, not necessarily the direction of current:

- **Why invert?**: We want inversion behavior in an inverting operational amplifier to flip signal polarity, eliminate common-mode voltage errors, and easily mix or attenuate signals

Common-mode voltage errors happen when an unwanted, identical voltage appears on both inputs of a differential measurement system, and the device fails to completely block it.

Flipping a negative voltage into a positive voltage allows standard analog-to-digital converters to process signals that cannot go below ground.

Because the inverting input is held at a steady virtual ground (zero volts), the input terminals experience no common-mode voltage fluctuations.

The virtual ground node allows multiple input signals to be safely added together through separate resistors without interacting with each other.

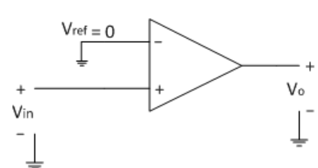
An inverting setup can reduce a signal's size (gain less than one), whereas a basic non-inverting setup cannot drop its gain below unity

- **The comparator circuit**: A comparator circuit compares two voltages and produces a binary-like output:
  - **HIGH** when the non-inverting input V_+ is greater than the inverting input V
  - **LOW** when V_+ is less than V_-.

  For the circuit shown, V_- is a fixed reference voltage and V_+ is the signal being tested:

$$V_{\text{out}} = \begin{cases} \textbf{HIGH} & V_+ > V_- \\ \textbf{LOW} & V_+ < V_- \end{cases}.$$

  The output normally approaches the comparator's supply rails. With a 0–5 V supply, LOW might be approximately 0 V and HIGH approximately 5 V. The exact voltages depend on the component.



- **hysteresis**: Hysteresis is a phenomenon where the state of a system depends on its history, meaning the output responds differently depending on whether the input is increasing or decreasing.

  In electronics and control systems, it introduces a intentional delay or "lag." Instead of having a single switching point, a system with hysteresis has two separate thresholds: an upper limit and a lower limit.

  The easiest way to understand hysteresis is to look at a household thermostat:
  - **Without hysteresis**: If you set your AC to 72°F, the system would turn on at 72.1°F and turn off at 71.9°F. If the room stays right around 72°F, your AC compressor would rapidly turn on and off every few seconds. This is called chattering, and it burns out motors quickly.

– **With hysteresis**: With Hysteresis: The thermostat introduces a built-in gap. The AC might turn on when the room warms up to 74°F (Upper Threshold), but it won't turn off until the room cools down to 70°F (Lower Threshold). The 4-degree difference is the hysteresis band, which keeps the system stable.

Hysteresis is implemented in electronic circuits by introducing positive feedback. By feeding a fraction of the output signal back to the non-inverting input (V) of an op-amp or comparator, the circuit dynamically shifts its own switching thresholds based on the current output state.

- $V_{\mathbf{sat}}$: The op-amp's output saturation voltage, the maximum magnitude its output can reach.

  When used as a comparator, the output normally settles near one of two levels:

$$V_{\text{out}} \approx +V_{\text{sat}},$$
$$V_{\text{out}} \approx -V_{\text{sat}},$$
$$.$$

  These are called positive and negative saturation.

- V_{OL}: Output-low voltage (V_{OL}) is the maximum voltage level measured at the output of a digital circuit when it is actively pulling the signal down to a logical "0" (low) state.

  V_{OL} is slightly above 0V (often between 0.1V and 0.6V) due to internal transistor resistance and stray voltage drops.

- V_{OH}: Output-high voltage (V_{OH}) is the minimum voltage level a digital circuit or integrated circuit (IC) guarantees to output when it is driving a logic "high" (1) state

- **Power supply vs input**: The power supply provides the energy the op-amp uses. The input provides the information that controls what the op-amp does with that energy.

  The supply pins connect the op-amp to an electrical energy source:

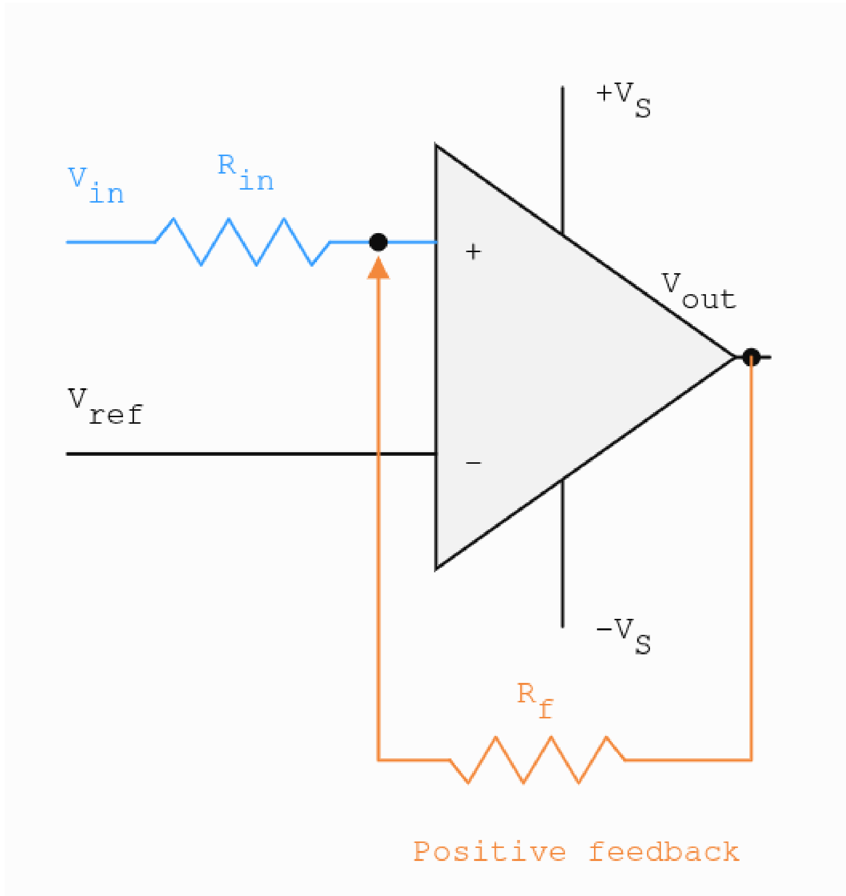
$$V_{S+} = +15V, \quad V_{S-} = -15V.$$

  Current from these connections powers the op-amp's internal transistors and supplies energy to the output.

  The supply rails also limit the output range:

$$-V_{\text{sat}} \leqslant V_{\text{out}} \leqslant +V_{\text{sat}}.$$

- **Non-inverting Schmitt trigger**: It is a comparator with positive feedback that produces two different switching thresholds.



This is an op-amp comparator + positive feedback.

When the output is low, say $-V_{\text{sat}}$, the negative output pulls the + input downward. The input must rise above the upper threshold V_{UT} before the + input exceeds the reference voltage. When this occurs, the output switches high.

V_{UT} and V_{LT} are the two input-voltage thresholds at which a Schmitt trigger changes its output state.

- V_{UT}: Upper threshold voltage
- V_{LT}: Lower threshold voltage

The upper threshold is used while V_{in} is increasing. Note that the resistor separates the voltages V_{in} and V_+, since

$$V_{\text{in}} - V_+ = I_{\text{R}_{\text{in}}} R_{\text{in}}.$$

Thus, they need not be equal. If there were no resistor and it were instead

$$V_{\text{in}} \to V_+,$$

resistance would be zero, so

$$V_{\text{in}} - V_+ = 0 \implies V_{\text{in}} = V_+.$$

The key distinction is that the low output pulls the op-amp's V_+ node downward, it does not normally pull the external source V_{in} downward. The resistor R_{in} separates those two voltages.

For the non-inverting Schmitt trigger:

$$V_{\text{in}} \to R_{\text{in}} \to V_+$$
$$V_{\text{out}} \to R_f \to V_+.$$

These two voltages compete to determine V_+.

We have,

$$I_{R_{\text{in}}} = \frac{\Delta V}{R} = \frac{V_{in} - V_+}{R_{\text{in}}},$$
$$I_{R_f} = \frac{\Delta V}{R} = \frac{V_{\text{out}} - V_+}{R_f}.$$

So, applying Kirchhoff's current law at V_+, we get

$$\frac{V_{in} - V_+}{R_{\text{in}}} + \frac{V_{\text{out}} - V_+}{R_f} = 0.$$

Recall that an op-amp used as a comparator examines

$$V_d = V_+ - V_-.$$

Its behavior is approximately

$$\begin{aligned} V_+ > V_- &\implies V_{\text{out}} = V_{OH} \\ V_+ < V_- &\implies V_{\text{out}} = V_{OL} \end{aligned}$$
$$.$$

Thus, the threshold from output low to output high is $V_+ = V_- = 0$, assuming that V_- is connected to ground, so $V_- = 0$.

So, when output is low, V_+ is driven down, when V_{in} is increased, V_+ rises, and when it rises above $V_- = V_{\text{ref}} = 0$, output switches.

With $V_+ = V_{\text{ref}} = 0$,

$$\frac{V_{in} - V_{\text{ref}}}{R_{\text{in}}} + \frac{V_{\text{out}} - V_{\text{ref}}}{R_f} = 0.$$

Thus,

$$V_{\text{in}} = V_{\text{ref}} + \frac{R_{\text{in}}}{R_f}(V_{\text{ref}} - V_{\text{out}}).$$

While the input is rising, the output is initially low:

$$V_{\text{out}} = V_{OL}.$$

Thus,

$$V_{UT} = V_{\text{ref}} + \frac{R_{\text{in}}}{R_f}(V_{\text{ref}} - V_{OL}).$$

If $V_{\text{ref}} = 0$,

$$V_{UT} = -\frac{R_{\text{in}}}{R_f} V_{OL}.$$

Because V_{OL} is negative, V_{UT} becomes positive.

Similarly, while the input is falling, the output is initially high, so

$$V_{LT} = V_{\text{ref}} + \frac{R_{\text{in}}}{R_f}(V_{\text{ref}} - V_{OH}).$$

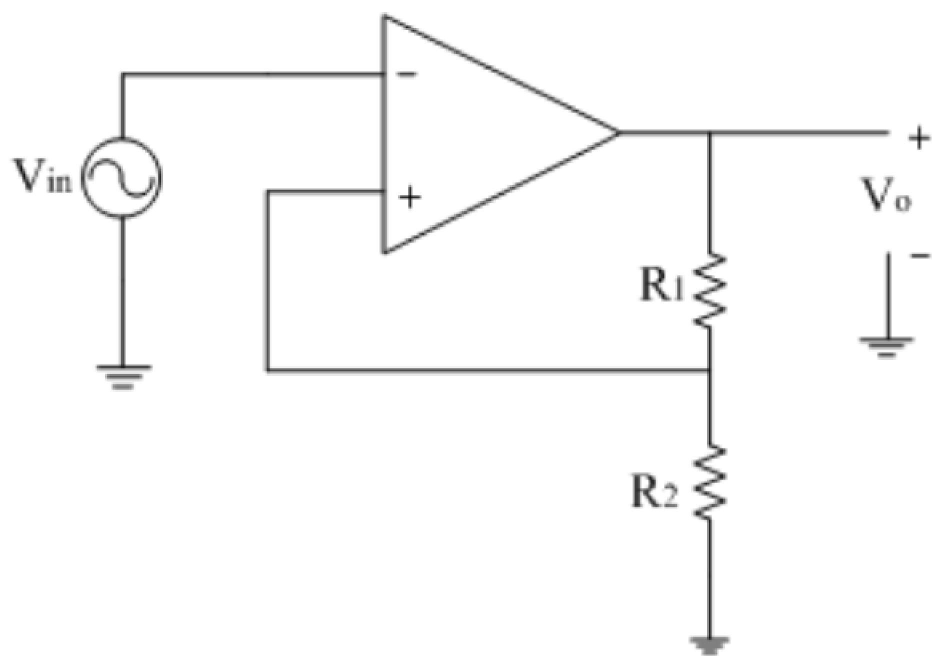
If $V_{\text{ref}} = 0$,

$$V_{LT} = -\frac{R_{\text{in}}}{R_f}V_{OH}.$$

- **Linear circuit**: A linear circuit is an electrical system where the output voltage and current are directly proportional to the input, creating a straight-line graph

  For example, if you double the input voltage, the output current also doubles.

- **Inverting Schmitt trigger**: An inverting Schmitt trigger is a comparator circuit with positive feedback. It converts a changing or noisy analog input into a clean HIGH or LOW output while using two different switching voltages.



Assume the reference point is $0V$. Consider the voltage divider connected to the non-inverting input. The current is the same through both resistors, that current is

$$I = \frac{\Delta V}{R} = \frac{V_{\text{out}} - 0}{R_1 + R_2} = V_{\text{out}}\frac{1}{R_1 + R_2}.$$

Notice that $V_+ - 0 = IR_2$, so

$$V_+ = V_{\text{out}}\frac{R_2}{R_1 + R_2}.$$

Define the feedback fraction

$$\beta = \frac{R_2}{R_1 + R_2}.$$

Thus,

$$V_+ = \beta V_{\text{out}}.$$

The important point is that V_+ depends on the current output state. When the output changes, V_+ changes with it.

If the output starts

$$V_{\text{out}} = V_{OL},$$

V_+ becomes negative, so the input V_- must be increased until the comparator threshold is reached. The upper threshold V_{UT} in this case is the voltage in which the output becomes LOW. In general,

$$V_{UT} = \beta V_{OH},$$
$$V_{LT} = \beta V_{OL}.$$

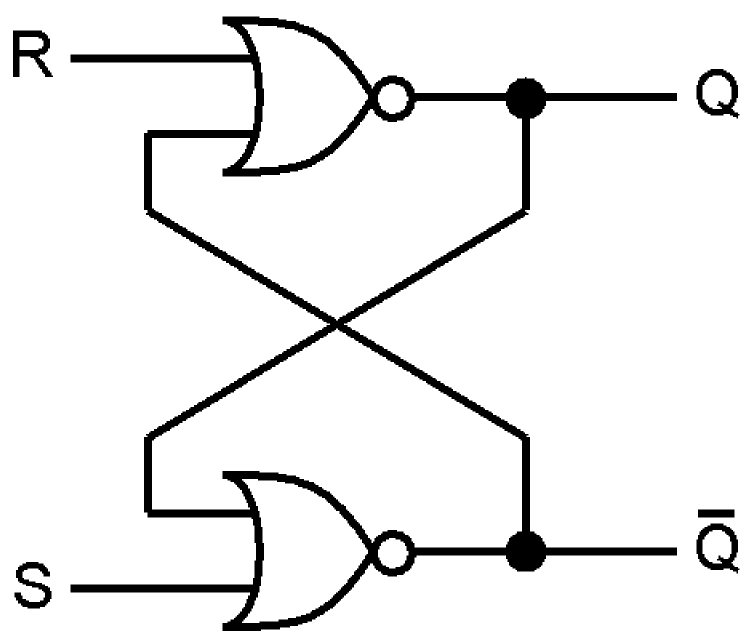
It is called an inverting Schmitt trigger because the input signal is connected to the op-amp's inverting input, marked $-$

The circuit's output changes in the opposite direction from the input:

- When V_{in} rises above the upper threshold V_{UT}, the output switches from HIGH to LOW.
- When V_{in} falls below the lower threshold V_{LT}, the output switches from LOW to HIGH.

This follows the comparator rule.

- **SR flip-flop**: The NOR implementation is the following



So,

$$Q_{n+1} = \overline{R + \bar{Q}_n} = \bar{R}Q_n,$$
$$\bar{Q}_{n+1} = \overline{S + Q} = \bar{S}\bar{Q}.$$

The characteristic table is

S	**R**	$\mathbf{Q_n}$	$\mathbf{Q_{n+1}}$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	undefined
1	1	1	undefined

Thus, Q_{n+1} is true when

$$\begin{aligned}\bar{S}\bar{R}Q_n + S\bar{R}\bar{Q} + S\bar{R}Q &= \bar{R}(\bar{S}Q + S\bar{Q} + SQ) \\ &= \bar{R}(\bar{S}Q + S(\bar{Q} + Q)) = \bar{R}(\bar{S}Q + S).\end{aligned}$$

Now, using the identity $x + yz = (x + y)(x + z)$,

$$\bar{S}Q + S = (S + \bar{S})(S + Q).$$

Thus,

$$\bar{R}(\bar{S}Q + S) = \bar{R}((S + \bar{S})(S + Q)) = \bar{R}(S + Q) = S\bar{R} + \bar{R}Q.$$

Now, the restriction for the SR flipflop is that $SR = 0$. So, $S = R = 1$ is forbidden. If $S = 1$, the restriction becomes $1R = R = 0$. Thus, $S = 1$ implies $R = 0$. So, we can say

$$S\bar{R} + \bar{R}Q = S + \bar{R}Q.$$

This is the characteristic equation for the SR flipflop. The restriction is that $SR = 0$. We have

S	**R**	**Operation**
0	0	Q_n
0	1	0
1	0	1
1	1	forbidden

Note: This version corresponds to **active-high**, cross-coupled NOR SR latch.

Notice that the NOR is active whenever

$$\overline{x+y} = \bar{x}\bar{y} = 1.$$

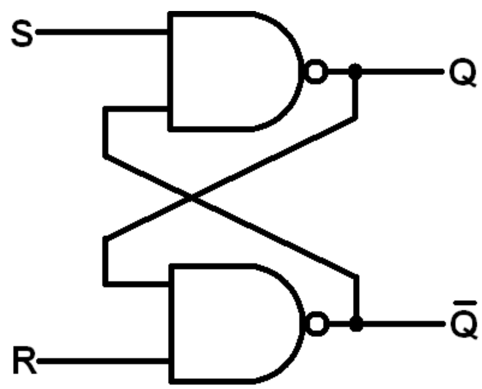
So, both inputs are off. Suppose $S = R = 0$, and $Q = 1$. Then,

$$SQ = 01 = 0 \implies \bar{Q} = 0,$$
$$R\bar{Q} = 00 = 1 \implies Q = 1.$$

Thus, we have the self-maintaining loop. When $S = 1$, it forces $\bar{Q} = 0$. This means the Q-producing gate now has both inputs low, assuming $R = 0$, so $Q = 1$.

When $R = 1$, $Q = 0$, so $SQ = 00 = 1 = \bar{Q}$.

- **Nand version**:



Notice

$$Q = \overline{\bar{S}\bar{Q}} = \bar{\bar{S}} + Q,$$
$$\bar{Q} = \overline{\bar{R}Q} = \bar{\bar{R}} + \bar{Q}.$$

For the nand gate,

$$\overline{xy} = \bar{x} + \bar{y}.$$

So, its active when either input is low. When $\bar{S} = 0$, $\bar{\bar{S}} = 1$, so the gate producing Q is forced to high. Now, if $\bar{R} = 0$,

$$\bar{Q} = 1 + 0 = 1.$$

Thus, $\bar{Q}$ is also high, so $S = R = 0$ is undefined in this version.

The characteristic table is

S	**R**	$\mathbf{Q_n}$	$\mathbf{Q_{n+1}}$
0	0	0	undefined
0	0	1	undefined
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Thus,

$$\begin{aligned} Q_{n+1} &= \bar{\bar{S}}\bar{R}\bar{Q} + \bar{\bar{S}}\bar{R}Q_n + \bar{S}\bar{R}Q_n = \bar{R}(\bar{\bar{S}}\bar{Q}_n + \bar{\bar{S}}Q_n + \bar{S}Q_n) \\ &= \bar{R}(\bar{\bar{S}}\bar{Q}_n + Q_n(\bar{\bar{S}}) + \bar{S}) = \bar{R}(\bar{\bar{S}}\bar{Q}_n + Q_n) \\ &= \bar{R}((Q_n + \bar{Q}_n)(Q_n + \bar{\bar{S}})) = \bar{R}(\bar{\bar{S}} + Q_n) \\ &= \bar{\bar{S}}\bar{R} + \bar{R}Q_n. \end{aligned}$$

Notice that since the condition is

$$\bar{S} + \bar{R} = 1,$$

when $\bar{S} = 0$, $\bar{R}$ is forced to 1. Thus, for $\bar{\bar{S}}\bar{R}$, when $S = 0$, $\bar{\bar{S}}\bar{R} = 1$, and when $S = 1$, $\bar{\bar{S}}\bar{R} = 0$. Hence, $\bar{\bar{S}}\bar{R} = \bar{\bar{S}}$ under the condition.

Therefore, the characteristic equation is

$$\bar{\bar{S}} + \bar{R}Q_n.$$

We have

S	**R**	**Operation**
0	0	undefined
0	1	1
1	0	0
1	1	Q_n

- **Latch**: A latch is a circuit that can store one bit of information:

  $$Q = 0 \quad \text{or} \quad Q = 1.$$

  It is called a latch because it "latches onto" a value and keeps that value even after the input that created it is removed.

Technically, the simple cross-coupled NOR or NAND circuit is an SR latch, not an edge-triggered flip-flop. A latch can respond as soon as its inputs change, whereas a flip-flop normally changes state only at a particular clock edge.

- **RC circuit**: An RC circuit is an electrical circuit made of a resistor (R) and a capacitor (C) connected together

  – A resistor R, which limits current.

  – A capacitor C, which stores electrical charge and energy.

  Its most important property is that the capacitor voltage changes gradually, rather than instantaneously.

  Consider a resistor and capacitor connected in series with a voltage source:

  $$V_{\text{in}} \to R \to C \to \text{GND}.$$

  The same current flows through both components. Kirchhoff's voltage law gives:

- **Back to the 555 timer**: